PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

FOR

B.Sc./B.Sc. (Hons)/B.Sc. (Hons. with research)
Discipline Specific Core Papers as per Table-2

FIRST YEAR
(SEMESTER SYSTEM)
EXAMINATIONS, 2024-25

(SEMESTER : FIRST AND SECOND)

i.e

First Semester: ***November/December, 2024***
Second Semester: ***April/May, 2025***

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Subject combinations offered in the Faculty of Science under NEP

Discipline Specific Courses (DSC)			
1	**2**	**3**	**4**
Physics	Chemistry	Mathematics	Bio-informatics
Botany	Computer Science and Applications	Zoology	Biotechnology
	Statistics		Industrial Chemistry
			Agriculture
			Fashion Designing
			Bio-chemistry
			Microbiology
			Microbial Food Technology

Note: Candidate will choose three subjects by selecting one course from each of the columns 1 to 4.

Table-2

NEP 2020 COLLEGES SCIENCE TEMPLATE: STUDENTS OPTING FOR MULTI-DISCIPLINARY/ INTER-DISCIPLINARY B.SC, B.SC HONOURS AND B.SCHONOURS WITH RESEARCH

Semester	Subject- A (4 Cr)	Subject-B (4 Cr)	Subject- C (4 Cr)	MDC /IDC (3 Cr)	AEC (2 Cr)	VAC (2 Cr)	SEC (3 Cr)	Internship/ Community Work (4 Cr)	Pbi/ HCP (2Cr)	Total
I	MAJ A-1	MAJ B-1	MIN C-1	MDC-1	Eng-1	VAC-1	SEC-1		Pbi-1/ HCP-1	24
II	MAJ A-2	MAJ B-2	MIN C-2	MDC-2	Eng-2	VAC-2	SEC-2		Pbi-2/ HCP-2	24
The student exiting the programme after one year, securing 48 credits, will be awarded UG Certificate provided he/she secures 4 credits from summerinternship/community work.										
The candidate will have to declare one of the three subjects as Minor in the 3rd semester. The other 02 subjects will remain his/her Major subjects										
III	MAJ A-3	MAJ B-3	MIN C-3 MIN C-4	MDC-3	Eng / MIL		SEC-3			24
IV	MAJ A-4 MAJ A-5	MAJ B-4 MAJ B-5	MIN C-5		Eng/ MIL	VAC-3				24
The student exiting the programme after two years securing 96 credits will be awarded UG Diploma provided he/she secures 4 credits from summer internship										
V	MAJ A-6 MAJ A-7	MAJ B-6 MAJ B-7	MIN C-6					Internship/ Community Work		24
VI	MAJ A-8 MAJ A-9	MAJ B-8 MAJ B-9	MIN C-7 MIN C-8							24
The student successfully securing 144 credits will be awarded BSc degree										
	36	36	32	09	08	06	09	04	04	144
VII	MAJ A-10 MAJ A-11	MAJ B-10 MAJ B-11	MIN C-9 MIN C-10							24
VIII	MAJ A-12 OR RM MAJ A – 13 MAJ A – 14 Or Dissertation of 08 credits	MAJ B-12 OR RM MAJ B – 13 MAJ B – 14 Or Dissertation of 08 credits								24
	56	56	40	09	08	06	09	04	04	192

Notes:-

1. In the first semester student is offered three different subjects A, B & C.
2. At the end of second semester each student has the option to choose any one of the two streams namely Single Major or Multi-Disciplinary. The student choosing Single Major stream is required to give his/her Major and Minor at end of the second semester.
3. The student choosing Multi-Disciplinary stream will continue with the three subjects and complete his/her three year degree by securing 36 credits each in the chosen two subjects as Major and 24 credits in Minor subject.
4. A student who secures 75% or more marks in aggregate till semester 6 (up to 3 year degree) shall be eligible for 4 YEAR DEGREE B.SC HONOURSwith research, on the condition that the Institution is a PU approved Research Centre
5. The student opting for Single Major must secure 50% credits of the total credits in the Major subject and obtain 24 credits from the Minor subject(s). To qualify for Minor in a particular subject, the student must attain at least 12 credits in that subject(s).
6. The student exiting the programme after one year securing 48 credits will be awarded UG Certificate provided he/she secures 4 credits from summer internship/community work. Further, the student exiting the programme after two year securing 96 credits will be awarded UG Diploma provided he/she secure 4 credits from summer internship. The student successfully securing 144 credits will be awarded BA degree.
7. Language paper chosen as MIL-1 and MIL-2 should be in the same subject.
8. Once Major and Minor papers are chosen in 3rd semester, the same will continue as Major and Minor subject in the subsequent semesters.
9. The candidate opting for Honours in research is required to study a paper of Research Methodology (RM) of 04 credits and write Dissertation of 08 credits in one of the two Major subjects.

Foreword

This document provides the syllabi for the first two semesters of various science subjects in the category of Discipline Specific Courses (DSC) for the first year students enrolled in four year degree programme in Science stream (except B.Sc. Biochemistery, B.Sc. Bioinformatics, B.Sc. Biotechnology, B.Sc Microbial and Food Technology and BCA) under NEP. The Syllabi of other categories i.e. MDC, AEC, VAC, SEC, Copulsory Punjabi and HCP have been given separately.

The University has approved two distinct templates leading to two types of degrees: Subject-Specific and Discipline-Specific. This document is alligned with the Template-2 which is Multidisciplinary Stream.

Advertising, Sales Promotion & Sales Management
Semester I

Course Title	**Fundamentals of Personal Selling**
Course Code	SAPDSC1
Typology of Course	**Subject A-1**
Course Duration/ LTP (Lectures, Tutorials, Practical)	**45 hours** of **lectures** for **Theory Course+ 30 hours of Practical** in one full Semester (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups of students (25 students in each group) will be created for the purpose of practical work for each unit of subject.**
Credits	4 Credits Theory + Practical = 3+1 credits
Max. Marks and Time	Max. Marks = 100 Time: 3 Hours 65 (Theory) + 10 (Internal Assessment) + 25 (Practical)
Course Objectives	In the Paper- Fundamentals of Personal Selling, Students will learn the theoretical and practical techniques used in selling goods and services in a Business – to-Business (B-to-B) environment. Emphasis is given to developing practical skills in presenting goods and services to prospective buyers.
Course Outcomes	Upon successful completion of this paper, students will have reliably demonstrated the ability to: a. Understand the concepts that are central to personal selling b. Understand the buying motives that is essential for creating successful sales. c. Develop and apply the various sales techniques needed to achieve a profitable sale. d. Develop and present an effective sales interview. e. Demonstrate effective sales presentation techniques
Instruction to Paper-Setter and candidates (Theory Paper):	Theory maximum Marks: 65 (10x2+3x15=65) Theory paper will be of 65 marks and maximum time allowed would be 3 hours. The candidate shall attempt 4 questions in all (one compulsory and one each from three units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 10. Each question will carry 2 marks.

Rest of the paper shall contain 3 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 15 marks.

For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in practical paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

Course Content

Unit-1 (15 Hours)

- Introduction to Personal Selling: Meaning, Definition & Characteristics, Nature and Importance of personal selling
- Buying Motives: Types of buying motives and their uses in personal selling
- Types of Selling Situations
- AIDAS Model of Selling

Unit-2 (15 Hours)

- Selling Process: Meaning, concepts and Process of effective selling, Prospecting and qualifying; Pre-approach; Approach; Presentation and demonstration; handling of objections and complaints; Closing the sale; techniques for closing the sale; Post sale activities, follow up and Dealing customer concerns and complaints.
- Qualities of successful sales person
- Types of Sales Personnel

Unit -3 (15 Hours)

- Sales Reports: Reports and Documents, Sales manual, Catalogue, Order Book, Cash Memo, Tour Diary, Daily and Periodical
- Reports
- Problems and difficulties in Selling
- Career opportunities in selling, Selling as an attractive career

Suggested Practical: (30 Hours)

- Submit any one assignment by each student at the end of session from following topics **(5 Marks):**
    a. Assignment on "Buying Motives with reference to any industry".
    b. Assignment on "Any Company's AIDAS Model of Selling".
    c. Assignment on "Analyse Selling Process of any company"
- Survey will be conducted by each student on Personal selling with reference to any company. Primary data will be collected and analysed in form of "survey-based project report on Personal selling". Questionnaire can be either developed by student or can be provided by concerned Department. **(15 Marks)**

- Viva-Voce of above submissions **(5 Marks)**. At the end of semester, the portfolios (Assignment and project Report) would be evaluated by the internal and external examiner. External Examiner will be designated by the Controller of Examinations, Panjab University. The Examiner would evaluate the portfolio for a maximum of 20 marks and conduct viva-voce examination for 5 marks. The award lists duly signed are to be sent the Controller of Examinations.

Pedagogy of the Course Work

Effective teaching learning process includes vital methods like audio-visual materials, class presentations, expert lectures, case studies and group discussion using ICT, Research Projects, Case Studies Discussions and Presentations, Experiential Learning via Simulation Exercises and Role Playing.

Suggested Readings

- Charles Futrell: Fundamentals of Selling, McGraw Hill.
- Still, Cundiff & Govani: Sales Management, Prentice Hall of India.
- Rusell, F. A. Beach and Richard H. Buskirk: Selling: Principles and Practices, McGraw Hill
- Futrell, Charles: Sales Management: Behaviour, Practices and Cases, The Dryden Press.

Semester II
Paper: Marketing Communication
Subject A-2

Course Title	**Marketing Communication**
Course Code	SAPDSC2
Typology of Course	**Subject A-2**
Course Duration/ LTP (Lectures, Tutorials, Practical)	**45 hours** of **lectures** for **Theory Course+ 30 hours of Practical** in one full Semester (1 credit = 15 hours of **theory learning** based on 3 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups of students (25 students in each group) will be created for the purpose of practical work for each unit of subject.**
Credits	4 Credits Theory + Practical = 3+1 credits
Max. Marks and Time	Max. Marks = 100 Time: 3 Hours 65 (Theory) + 10 (Internal Assessment) + 25 (Practical)
Course Objectives	The Objective of this paper is to provide insights into the communication aspects of marketing. They will be able to understand the steps involved in development of effective marketing communication and determining the various tools of promotion.
Course Outcomes	Upon successful completion of this paper, students will have reliably demonstrated the ability to: a. Explore the fundamental concepts of marketing communication and learn effective communication. b. Develop effective marketing communication simulations. c. Categorize the promotion mix tools to be used for creating public relations, sales promotion and direct marketing. d. Conversant with the process of integrated marketing communication process and also how to manage cultural diversity through communication
Instruction to Paper-Setter and candidates (Theory Paper):	Theory maximum Marks: 65 (10x2+3x15=65) Theory paper will be of 65 marks and maximum time allowed would be 3 hours. The candidate shall attempt 4 questions in all (one compulsory and one each from three units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 10. Each question will carry 2 marks.

Rest of the paper shall contain 3 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 15 marks.

For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in practical paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

Course Content

Unit-1 (15 Hours)

- Communication: Meaning, Nature and importance of communication, Types of Communication, objectives of Communication, Communication Process, Problems in Communication and suggestions for making communication effective.

Unit-2 (15 Hours)

- Marketing Communication: Meaning, Types, factors, Process of Marketing Communication, Steps to develop effective Marketing Communication, Noise in Marketing Communication, Application of the Communication process in Marketing

Unit-3 (15 Hours)

- Promotion or Communication Mix: Meaning, components- Advertising, Personal Selling, Public Relations, Sales Promotion, their meaning and distinct characteristics; factors, Process, Optimum Promotion Mix, Advertising Vs. Sales Promotion, Advertising Vs. Personal selling
- IMC (Integrated Marketing Communication): Integrated marketing Communication: the target audience, determining objectives, designing the message, Selecting Communication Channel, establishing budget, Promotional Tools, Factors in setting the Marketing Communication Mix and Measuring Results.

Suggested Practical: (30 Hours)

- Submit any one assignment by each student at the end of session from following topics **(5 Marks):**
    a. Assignment on "Role of Communication in marketing".
    b. Assignment on "Problem faced in Marketing Communication by any company in form of discussion of case study related to any company".
    c. Assignment on "Design and develop Integrated promotion campaign for rural market."

- Survey will be conducted by each student on Promotion Mix with reference to any company. Primary data will be collected and analysed in form of Project report on Promotion Mix". Questionnaire can be either developed by student or can be provided by concerned Department. **(15 Marks)**
- Viva-Voce of above submissions **(5 Marks)**. At the end of semester, the portfolios (Assignment and Research Report) would be evaluated by the Internal and external examiner. External Examiner will be designated by the Controller of Examinations, Panjab University. The Examiner would evaluate the assignment and project report for a maximum of 20 marks and conduct viva-voce examination for 5 marks. The award lists duly signed are to be sent the Controller of Examinations.

Pedagogy of the Course Work

Effective teaching learning process includes vital methods like audio-visual materials, class presentations, expert lectures, case studies and group discussion using ICT, Research Projects, Case Studies Discussions and Presentations, Experiential Learning via Simulation Exercises and Role Playing.

Suggested Readings

- E Belch, George; Michael A Belch; KeyoorPurani: Advertising and Promotion: An Integrated Marketing Communications Perspective (SIE), McGraw Hill Education
- Terence A. Shimp, J. Craig Andrews: Advertising, Promotion, and Other Aspects of Integrated Marketing Communications, South West Cingage Learning
- Shah, Kruti: Advertising and Integrated Marketing Communications, McGraw Hill Education

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Agriculture

Semester-I

Course Title : ELEMENTS OF AGRONOMY AND AGRICULTURE
Course Cod : AGRDSC1
Objective: To give students elementary information on basic agronomic principles and practices.

Max. Marks: 100 **Course Credits:** 3+1 = 4

1.	**Theory: 75**	**Written Exam: 60**	**Internal Assessment: 15**
2.	**Practical: 25**	**Written Exam: 20**	**Internal Assessment: 5**

Periods per week:

1. Theory - Four of 45 minutes duration (Three Hours)
2. Practical - One of two hours duration (Two Hours)

INSTRUCTIONS FOR PAPER SETTER:

Students will attempt five questions out of a total of nine. Question No. 1 will be COMPULSORY and comprise 8 short answer-type questions of one and a half marks each and will cover the whole of the syllabus. Units 'I', 'II', 'III', and 'IV' will have two questions each, and one question is to be attempted out of each Unit.

Teaching Aids: Teachers can use lecture, lecture cum demonstration, project, assignments, problem-solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to the black/whiteboard.

Unit-I

- Evolution of agriculture, ancient agriculture, history of Indian Agriculture, Principles of agronomy, classification of crops.
- Crop rotation: principles, limitation, advantages; rotational intensity, cropping scheme, cropping intensity.
- Cropping system: intercropping, mixed cropping, multiple cropping, and relay cropping.

Unit-II

- Agronomy as a science and its relationships with other sciences. Germination, maturity, harvesting, and storage of crop plants.
- Physical & biological properties of soil, soil organic matter.
- Tillage principles, tillage practices for different soil types and crops.

Unit-III

- Seedbed preparation. Characteristics of good seedbeds.
- Methods of sowing and their suitability under different conditions.
- Seedling practices in relation to types of seeds, time of sowing, and soil moisture.

Unit-IV

- Weed characteristics, dissemination and competition for growth factors, and problems caused by them.
- Common weeds, losses, and benefits, crop-weed associations.
- Common methods of weed control.
- Green manuring – advantages, limitations, green manure crops.

Practical:

- Land measurement
- Identification and practices in the use of farm hand tools, bullock and tractor-driven implements.
- Identification of crop seeds, crop plants, and important weeds.
- Knowledge of various fertilizers – collection, weeding, hoeing in various crops.
- Collection of weeds
- Use of spray pumps.
- Calculation of herbicide doses for different crops.

Suggested Readings:

1. Reddy S.R., Principles of Agronomy. Kalyani Publishers, Ludhiana, 2018.
2. Handbook of Agriculture, I.C.A.R. Publications, New Delhi, 2008.
3. Weeds of North India, I.C.A.R. Publications, New Delhi, 2008.
4. Package of Practices for Rabi and Kharif crops, P.A.U. Publications, Ludhiana, Corresponding year.

Semester: II

Course Title: BASICS OF AGRICULTURAL ECONOMICS
Course Code: AGRDSC2
Objective: To give students elementary information on basic principles of economics related to agriculture.

Max. Marks: 100 **Course Credits:** 3+1 = 4

1.	**Theory: 75**	**Written Exam: 60**	**Internal Assessment: 15**
2.	**Practical: 25**	**Written Exam: 20**	**Internal Assessment: 5**

Periods per week:

1. Theory - Four of 45 minutes duration (Three hours)
2. Practical - One of two hours duration (Two hours)

INSTRUCTIONS FOR PAPER SETTER:

Students will attempt five questions out of a total of nine. Question No. 1 will be COMPULSORY and comprise 8 short answer-type questions of one and a half marks each and will cover the whole syllabus. Units 'I', 'II', 'III', and 'IV' will have two questions each, and one question is to be attempted out of each Unit.

Teaching Aids: Teachers can use lecture, lecture cum demonstration, project, assignments, problem-solving, inductive and deductive methods, and can use L.C.D., L.E.D., Multimedia projector in addition to the black/whiteboard.

Unit-I

- Agricultural Economics: Meaning, scope, and its relationship with other social and agricultural sciences.
- Resource Management in Agriculture: land, labor, and capital.
- Agriculture and Economic Development: Introduction, role, share of agriculture in economic development; Role of agriculture in the economic development of India and its comparison with other countries.
- Factors affecting agricultural development.

Unit-II

- Risk and Uncertainty: Introductions, risk types and risk as a choice indicator; different theories of risk and uncertainty; measures to minimize risk and uncertainty.
- System of Farm Organization: Introductions, types, criteria for choice in India, and suitability of alternate systems.
- Farm Size and Practices: Holdings and farm.
- Dynamics of Population: Population growth and its effect on the agricultural growth rate.

Unit-III

- Agricultural marketing, efficient marketing system, classification of markets, problems of Indian agricultural marketing, and measures to improve agricultural marketing.
- Rural Indebtedness: introduction, extent, nature, and causes of rural indebtedness, Government measures to reduce rural indebtedness.
- Agricultural Finance.
- Agricultural credit institutions: structure of agriculture credit system, multi-agency approach, regional rural banks, NABARD, ARDC.

Unit-IV

- Self-Help groups, cooperative credit structure - three-tier structure.
- Rural Poverty: Introduction, factors responsible for poverty, suitable measures to reduce poverty, and poverty alleviation program.
- Cottage and small industries: Features and classification of cottage industries, difference between cottage and small industries, role, and problems of cottage and small industries.
- Different measures to promote SSI and the role of Government, small industries development bank of India and its role, khadi and village industries (KVIC).

Practical:

- Calculations of interest.
- Calculations of depreciation.
- Farm Budgeting.
- Preparation of balance sheets and other final statements of financial importance for agriculture farms.
- Study tours of Marketing Centers of the local area.

Suggested Readings:

1. Lekhi, R.K. and Singh, J., Agricultural Economics, Kalyani Publishers, Ludhiana, 2016.
2. Black, J.D., Introduction of Economics for Agriculture, Fromount Pierre National Press, 1955.
3. Nanavati, M.B. and J.J. Anjaria, The Indian Rural Problem. The Indian Society of Agricultural Economics, Bombay, 1944.
4. Memoria, C.B. and B.B., Agricultural Problems in India, Kitab Mahal, Allahabad, 2007.

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BIO-CHEMISTRY

Multidisciplinary Scheme

Semester I

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:
1. Total No. of questions will be nine. All questions carry equal marks.
2. Q. No. 1 will be compulsory. It will consist of short questions covering the entire syllabus
3. Besides question Number 1, there will be 4 sections of 2 questions each.
4. All other questions may contain 2-3 parts.
5. Questions should be uniformly spread over the entire syllabus.
6. Students will be required to attempt 5 questions in all including Q. No. 1 and at least one question from each of the 4 sections.

BCHDSC1 : Biomolecules I (3+1) Marks: 75 (60 + 15 Int.)

Objective: To learn about biomolecules, their structure and functional significance.

SECTION-I

(Lectures: 10)

Introduction to biochemistry and its scope. Water: physical properties, as a biological solvent and structure of water, dissociation of water. pH and pOH, buffer solution. Henderson Hassel Balch equation, acid-base indicators, buffers and physiologically important buffers, dialysis and osmosis.

SECTION-II

(Lectures: 10)

Definition and classification of carbohydrates. Fischer and Haworth structures of carbohydrates. Stereoisomerism andmutarotation. Anomeric forms of monosaccharides. Derivatives of monosaccharides (glycosides, deoxysugars, amino sugars and other derivatives of biological importance). Oligosaccharides of biological importance (structure of maltose, lactose, sucrose, cellobiose, trehalose, and raffinose). Characteristic reactions of monosaccharides: reactions with hydrazine, hydrogen cyanide, hydroxylamine; reduction and oxidation of sugars; Homo-and hetero-polysaccharides (structures of amylose, amylopectin, starch, glycogen, cellulose, chitin). Hetero-polysaccharides) as components of connective tissue. Polysaccharides of bacterial cell wall.

SECTION-III

(Lectures: 10)

Definition and classification of fatty acids (saturated and unsaturated). Essential fatty acids. Important reactions of functional groups present in fatty acids. Characteristics of fatty acids and fats (saponification, iodine, acid, acetyl and peroxide values). Refractive index, melting point, boiling point and their relation to molecular size. Properties of glycerol. Fats as source of energy. Waxes. Structures, characteristics and functions of lipids: triacylglyceroland phospholipids such as lecithin (Phosphatidylcholines), lysolecithins, cephalins (phosphatidylethanolamines), phosphatidylserines, Phosphatidylinositol, sphingomyelins, plasmalogens, cerebrosides, gangliosides, sulfatides.

SECTION-IV

(Lectures: 10)

Lipoproteins– composition, classification and biological functions. Liposomes. Terpenes and steroids – Terpenes of biological significance e.g., carotenes, phytol. Cholesterol and other animal sterols. Color reactions of sterols. Sterols of yeast and fungi (mycosterols). Phytosterols. Steroidal hormones. Bile acids. Structure and properties of eicosanoids-prostaglandins, leukotrienes, thromboxane, prostacyclins

PRACTICALS:

MAJ A-1-BCH/MAJ B-1-BCH/MIN C-1-BCH:Biomolecules I **Marks: 25 (20 +5 (Int.)**

1. Qualitative tests for carbohydrates.
2. Determination of achromatic point of saliva
3. Qualitative tests for Lipids/Fats.
4. Estimation of saponification value of fat/oil.
5. Estimation of iodine value of fat/oil.
6. Isolation of starch from potato/gluten from wheat.
7. Demonstration of Osmosis and Dialysis

Suggested Books:

1. Nelson D.L.Cox M M : Biochemistry, Worth Publishers Inc., New York. (7th Edition, January 2017)
2. Conn E.E & Stumpf. P.K. : Outlines of Biochemistry, John Wiley & Sons, U.S.A.
3. Zubays Principles of Biochemistry :(**Veer Bala Rastogi**) 2018, Meditech Publication
4. Robert K. Murrary P.A. : Harper's Illustrated Biochemistry, Prentice Hall Int., D.K. Granner and V.W. London (34th Edition, 2018). Rodwell.
5. Dean R. Appling (Author), Biochemistry : Concepts and Connections, Pearson publisher (2016), Spencer J. Anthony-Cahi

Semester II

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS:
1. Total No. of questions will be nine. All questions carry equal marks.
2. Q. No. 1 will be compulsory. It will consist of short questions covering the entire syllabus
3. Besides question Number 1, there will be 4 sections of 2 questions each.
4. All other questions may contain 2-3 parts.
5. Questions should be uniformly spread over the entire syllabus.
6. Students will be required to attempt 5 questions in all including Q. No. 1 and at least one question from each of the 4 sections.

BCHDSC2: Biomolecules II (3+1) Marks: 75 (60 + 15 Int.)

SECTION–I

(Lectures: 10)

Amino acids: common structural features. Stereoisomerism and RS system of designating optical isomers.
Classification based on the nature of "R" groups. Amino acids present in proteins and non-protein amino acids.Specialized role of amino acids. Physical and chemical properties of amino acids. Titration of amino acids. Rigid and planar nature of a peptide bond. Biologically active peptides.

SECTION–II

(Lectures: 10)

Proteins: Structural levels Primary, Secondary, Tertiary and Quaternary structures of proteins structure. Folding of peptide chains into regular repeating structures (α helix, β-pleated sheets). β -turn in polypeptides. Forces stabilizing structure and shape of proteins. Native proteins and their conformations. Behaviour of proteins in solutions. Salting in &salting out of proteins. Denaturation of proteins. Structural and functional diversity of proteins, fibrous proteins (keratins, collagen& elastin), globular proteins (hemoglobin, myoglobin) and conjugated proteins.

SECTION-III

(Lectures: 10)

Nucleic Acids: structure and properties of purine and pyrimidine bases. Nucleosides and nucleotides. Biologically important nucleotides. Double helical model of DNA and forces responsible for it. Different forms of DNA (A, B and Z type), denaturation of DNA. Physical and chemical properties of nucleic acids. Chemical and enzymatic hydrolysis of nucleic acids. Types of RNA and ribozyme.

SECTION-IV

(Lectures: 10)

Porphyrins: porphyrin nucleus and classification of porphyrins. Metalloproteins, Heme and other metalloporphyrins occurring in nature. Detection of porphyrinsspectrophotometrically and by fluorescence. Chemical nature and physiological significance of bile pigments.

Suggested Books:

1. Nelson D.L.Cox M M : Biochemistry, Worth Publishers Inc., New York. (7th Edition, January 2017)
2. Conn E.E&Stumpf. P.K. : Outlines of Biochemistry, John Wiley & Sons, U.S.A.
3. Zubays Principles of Biochemistry : (**Veer BalaRastogi**) 2018, Meditech Publication
4. Robert K. Murrary P.A. : Harper's Illustrated Biochemistry, Prentice Hall Int., D.K. Granner and V.W. London (34th Edition, 2018). Rodwell.
5. Dean R. Appling (Author), Biochemistry : Concepts and Connections, Pearson publisher (2016), Spencer J. Anthony-Cahi

PRACTICALS:

MAJ A-2-BCH/MAJ B-2-BCH/MIN C-2-BCH:Biomolecules II Marks: 25 (20 +5 (Int.)

1. Qualitative tests for amino acids
2. Titration curve of an amino acid and determination of its pKa and isoelectric pH.
3. Separation of amino acids by thin layer chromatography
4. Ammonium sulphate fractionation of serum proteins.
5. Absorption Spectra of DNA and Proteins by UV visible spectroscopy
6. Estimation of Protein by Biuret /Lowry Method

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Bioinformatics

Semester-I

BIOINFORMATICS-I

(Credit: 4)
Theory: 60
Int. ass: 15
Time: 3 Hours

Objective:
Introduction to Biomolecules exposes the students to study structure and properties of biomolecules and their related bioinformatics resources.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Nucleic acids:
Nitrogenous Bases, nucleosides and nucleotides.

Overview of DNA structure: Watson and Crick DNA double helix model, Conformations of DNA: A-DNA, B-DNA & Z-DNA.

Overview of RNA: types and roles of different RNAs: mRNA, rRNA, tRNA, snRNA and snoRNA.

Amino acids: Standard amino acids and the general properties.

Classification and structures: non-polar, polar but uncharged and charged amino acids.

Introduction to Proteins: Overview ofProtein Structure.

Primary structure - definition, peptide bond, dihedral angles, and Ramachandran plot.

Secondary structure - definition, alpha helix, beta sheet, and beta turn.

Tertiary structure - definition, interactions stabilizing tertiary structure.

QuaternaryStructure - definition, interactions stabilizing quaternary structure

UNIT – II

Introduction to Bioinformatics: Definitions, History, Goal, Scope and its Applications.

Literature Database: PubMed.

Biological Databases Classification: Primary, Secondary and Specialized databases.
Nucleotide Sequence Database: Genbank, EMBL, INSDC, Indian Biological Data Centre (IBDC) and its data portals.
Protein Sequence Database: Uniprot.
Database flat file formats: Genbank/Genpept, EMBL and Uniprot flat files.
Sequence file formats: FASTA

Sequence Alignment: Evolutionary basis of Alignment, Concept of Sequence Homology, Similarity, and Identity. Applications of Sequence Alignments.

Pairwise sequence alignment: Concept of Global and Local alignment, Dot Matrix method, Dynamic programming (Needleman Wunsch algorithm and Smith-Waterman algorithm) Concepts behind General gap and affine gap penalty.

Recommended Books:

1. Donald Voet and Judith G. Voet (2010) Biochemistry, 4th Edition, Wiley Publisher.
2. David L. Nelson and Michael M. Cox (2017). Lehninger Principles of Biochemistry, 7th Edition, WH Freeman Publisher.
3. Reginald H. Garrett and Charles M. Grisham (2023) Biochemistry, 7th Edition, Brooks/Cole Publisher.
4. Introduction to Bioinformatics by V. Kothekar, 1st Edition, (2006), Dhruv Publications, Delhi.
5. Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
6. Bioinformartdics: A practical guide to the analysis of genes and proteins. Ed. By Baxevanis, 3rd Edition, (1998) John Wiley & Sons, Inc. Publication.

BINF-E1051: BIOINFORMATICS-I(PRACTICAL)

Max Marks: 25 (20+5)

1. Draw the amino acids using Chemsketch.
2. Plotting of Ramachandran map of a protein.
3. Visualization of DNA using Pymol.
4. Searching literatures over PubMed.
5. Understanding and using Sequence information resources on web: EMBL-ENA,Genbank, EMBL, IBDC and Uniprot.
6. The information retrieval using Gquery.
7. Performing Sequence alignment using DOT PLOT method

Semester-II

BIOINFORMATICS-II

(Credit: 4)
Theory: 60
Int. ass: 15
Time: 3 Hours

Objective:
This subject offers various bioinformatics tools that helps in analysis of biological sequence data and phylogenetic analysis of proteins and nucleotides by using various computational tools.

General Instruction:

- The question paper will have seven questions, and each question have 12 marks. The first question would be compulsory having sub-parts covering the entire syllabus in the form of short and objective type questions.
- The remaining six questions will be set out of the Units I and II consisting of three questions from each unit.
- A candidate is required to attempt five questions in all by selecting two questions, from each unit and the first compulsory question.

UNIT – I

Tools for Pairwise Sequence Alignment:BLAST: Algorithm, types, understanding the BLAST output and the significance of E-value and Bit score, different uses of BLAST.

FASTA algorithm, Difference between BLAST & FASTA.

Substitution matrices: Amino acid substitution matrices (PAM and BLOSUM)

Multiple Sequence Alignment (MSA): SP (Sum of Pairs) measure to determine scoring function, Multiple Sequence Alignment methods: Progressive (Clustal Omega and PILEUP), Iterative and Block-based methods of MSA. Application of MSA.

Tools for MSA: Clustal Omega, Clustal-X and T-coffee.

UNIT – II

Phylogenetic Analysis: Basic terminology in Phylogenetics, Introduction to phylogenetic analysis and its application, Phylogenetic tree terminology and topologies.

Methods of Phylogenetic Tree Construction:
Distance Based Methods: NJ, FM & UPGMA
Character Based Methods: Maximum Parsimony & Maximum Likelihood.
Phylogenetic Tree Evaluation Methods: Bootstrapping & Jackknifing.

Tools for Phylogenetic Analysis: Introduction to Phylip and MEGA packages.

Protein Structural databases:PDB, SCOP and CATH

Recommended Books:

- Bioinformatics: A modern approach by Vittal R. Srinivas, Prentice/ Hall of India Pvt. Ltd. New Delhi, (2005)
- Bioinformatics by David Mount, Cold Spring Harbor Laboratory, (1989).
- Bioinformatics: Ardpractical guide to the analysis of genes and proteins. Ed. By Baxevanis 3rd Edition, (1998) John Wiley & Sons, Inc. Publication
- BLAST by Ian Korf, Mark Yandell, Joseph Bedell, O'Reilly, 1st Edition, (2003).
- Molecular Evolution: a phylogenetic approach, (1998), 352p, ROM and Holmas FC, Blackwell Science.

BINF-E2051:BIOINFORMATICS-II (PRACTICAL)

Max Marks: 25 (20+5)

1. Performing BLASTp/n and interpreting its results.
2. Performing Multiple Sequence Alignment using Clustal Omega
3. Building Phylogenetic tree using:
   a) Distance-based & character-based methods by PHYLIP or MEGA
   b) Tree evaluation
4. Understanding the PDB and its file format
5. Downloading and visualizing protein structure from PDB

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Biotechnology

Semester-I

BTEM1- Fundamentals of Biotechnology –I

(3+1 Credits)

Objectives: -

This course will introduce the basic concepts of biotechnology to the students. They will learn about the history of biotechnology; the foundations of modern biotechnology; the role of biotechnology in fermentation industry, environment and modern medicine and the ethical implications of biotechnology.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs/week)
(1 credit = 30 hours of practical based on 2 hrs/week)
(1 Semester = 15 weeks)

Credits: 4 credits (Each credit is equal to 25 marks)

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Unit-I

Advent, scope and basics of biotechnology.Green, White, Red and Blue biotechnology.
Study of model organisms: *E. coli*, *Saccharomyces cerevisiae*, *Neurosporacrasa*, *Drosophila melanogaster, Caenorhabditiselegans, Mus musculus*
Role of viruses and bacteriophages in biotechnology.

Unit-II

Structure and function of the cell: the basic unit of life
Morphology of the cell: Differences in Prokaryotic and eukaryotic Cell,
Cell types including cellular specialization and differentiation.

Unit-III

Introduction to basic techniques like sterilization, centrifugation, electrophoresis, chromatography, sonication

Biosafety management: Key to the responsible use of biotechnology, Biosafety levels, Goodsafetypractices.

Introductiontopatentsinbiotechnology.

Unit-IV

Genetically modified organisms (GMOs) - transgenic plants and animals and theirapplications in biotechnology.Public concerns and risks associated with genetic engineering

Biotechnology Research in India. Biotechnology Institutions in India (Public and Private Sector) their objectives and achievements.

Ethical and social issues in biotechnology.

SUGGESTED READING:

1. Comprehensive Biotechnology, Murray Moo-Young, 2nd edition (2011), Pergamon Press.
2. Introduction to Biotechnology, William J. Thieman& Michael A. Palladino, 3rd edition (2012), Benjamin Cummings.
3. Biotechnology Expanding Horizons, BD Singh, 4th edition (2012), Kalyani Publishers.
4. The Ethics of Biotechnology (Biotechnology in the 21st Century), Jonathan Morris, Chelsea House Pub (L), 1st edition (2005).
5. Biotechnology, Applying the Genetic Revolution, David P. Clark & Nanette J. Pazdernik, 1st edition (2008), Academic Cell.
6. Molecular Biotechnology, Sandy B. Primrose, 2nd edition (1991), Blackwell Scientific Publications, ISBN.17
7. Biotechnology: Demystifying the concepts, David Bourgaize, Thomas R. Jewell & Rodolfo G. Buiser,1st edition(1999), Benjamin Cummings.

BIOT-MIN 1 (Fundamentals of Biotechnology-I) (Practical) Credits:1

1. Good laboratory practices followed in biotechnology laboratory (sterility
2. Introduction and use of equipments like electrophoretic apparatus, spectrophotometer and centrifuges
3. Concept of chemical hoods.
4. Concept of laminar flow hoods and Biosafety cabinets: principle and working.
6. Concept of sterilization: autoclave (principle and working).
7. Handling and disposal of hazardous reagents (acids, carcinogenic chemicals like acrylamide, ethidium bromide)

Semester-II

BTEM2- Fundamentals of Biotechnology –II

(3+1 Credits)

Objectives: -

This course will introduce the basic concepts of biotechnology to the students. Microbes play significant role in understanding Biotechnology industry so study of microbes frombasic to advance level, with understanding of biochemistry, cell structure and application makes this paper significant.

Course Duration

45 hours of lectures for Theory + 30 hours of practical
(1 credit = 15 hours of theory learning based on 3 hrs/week)
(1 credit = 30 hours of practical based on 2 hrs/week)
(1 Semester = 15 weeks)

Credits: 4 credits (Each credit is equal to 25 marks)

External Theory	Internal Theory	External practical	Internal practical
68	07	22	03

Theory Paper: The exam of this course will be of 3 hours duration.

Instructions for paper setters and candidates:

The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (COMPULSORY) of short answer type covering the whole syllabus. The students are required to attempt one question from each unit.

Compulsory question will be of 12 marks. Rest all questions will be of 14 marks each

Credits: 3

UNIT-I

Introduction to microbiology: History A.Leewenhook,L.Pasteur,R.Koch,J.Lister, Alexander Fleming.
Techniques for microbial identification and classification.
Principle of microscopy: Bright field, dark field, phase contrast, fluorescent, electron microscopy.

UNIT-II

Types of culture media, Staining techniques, Isolation of pure cultures**.**Maintenance and Preservation of microbial cultures.
Morphology of bacteria, with major emphasis on bacterial cell wall. Gram positive and Gram negative bacteria. The bacterial endospore.
Products from microorganisms: Primary and Secondary metabolites and their commercial uses.

UNIT-III

Water: Physico chemical properties of water, dissociation and association constants. pH and buffers.
Carbohydrates: Structure of important mono, di-, oligo- and polysaccharides, glycoproteins, peptidoglycan,glycolipids and lipopolysaccharides. Storage polysaccharides and cell walls.Epimers, Isomers, pyranose and furanose forms, Reducing and nonreducing sugars.
Nucleic Acids: Structures of RNA & DNA

UNIT-IV

Proteins: Classification of proteins according to biological functions (Enzymes, transport, storage. Contractile, structure, defense and regulatory), Classification, structure and chemical reactions of amino acids, primary, secondary, tertiary and quaternary structure in proteins.

Lipids: Classification of lipids and fatty acids, general functions and structure of major lipid subclasses, acylglycerols, phosphoglycerols, phosphoglycerides, sphingolipids, glycosphingolipids.

Reference Books:

1. Tortora, G.J., Funke, B.R. and Case, C.L. (2009) Microbiology: An introduction (Benjamin/ Cummings publishing company, Inc).
2. R. Y. Stanier, M. Doudoroff, E. A. Adelberg (1999). General microbiology (MacMillian press London).
3. M.J. Pelczar, E.C. Sun Chan, N.R. Krieg (1986). Microbiology (Tata McGraw Hill Publication, New Delhi).
4. H.G. Schlegel, C.Zaborosch, M. Kogut (1993).General microbiology (Cambridge University Press).
5. S.C. Prescott, C.G. Dunn (1959). Industrial microbiology (McGraw- Hill).
6. Lehninger A.L., Nelson D.L. and Cox M.M. (2005).Principles of biochemistry (W. H. Freeman, USA).
7. Stryer L, J. M. Berg, J.L. Tymoczko (2001). Biochemistry (W.H. Freeman and Company, New York).
8. Rawn J.D. (1989). Biochemistry (Neil Patterson Publishers).

BIOT-MIN 2 Fundamentals of Biotechnology –II (Practical)

1. Preparation of media, cotton plugging and sterilization.
2. Isolation of microorganisms from air and soil samples.
3. Dilution and pour plating techniques.
4. Gram staining, spore staining
5. Preparation of physiological buffers.
6. Estimation of carbohydrates in given solution by Anthrone method.
7. Methods of protein estimation.
8. Methods for Lipid estimation.

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BOTANY

SEMESTER–I

Course Title: PLANT DIVERSITY-I **Course Code: BOTDSC-1**

Objective:
The basic objective of this paper is to make students aware of the diversity in life forms of the plant kingdom. A systematic study of algae, fungi, and bryophytes would familiarize students with the structural differentiation and heterotrophic and autotrophic modes of nutrition in the plant kingdom. This paper will form the basis of advanced study in Botany.

Max. Marks: 100 **Course Credits: 3 + 1 = 4**

1. Theory: 75	Written Exam: 60	Internal Assessment: 15
2. Practical: 25	Written Exam: 20	Internal Assessment: 5

Periods per week:
1. **Theory:** Four of 45 minutes duration (Three hours)
2. **Practical:** One of two hours duration (Two hours)

Instructions for Paper Setters:
Students will attempt five questions out of a total of nine. Question No. 1 will be COMPULSORY and comprise 8 short-answer type questions of one and a half marks each and will cover the whole of the syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem-solving, inductive and deductive methods, and can use L.C.D., L.E.D., multimedia projector in addition to black/whiteboard.

UNIT–I

Algae: General Characters; systematic position, structure, and life history of Oscillatoria (Cyanophyceae), Volvox (Chlorophyceae), Ectocarpus (Phaeophyceae), Batrachospermum (Rhodophyceae) and economic importance of algae.

UNIT–II

Fungi: General characters; systematic position, structure, and life history of Albugo candida (White rust of crucifers), Rhizopus, and Saccharomyces.

UNIT–III

Systematic position, structure, and life history of Puccinia graminis-tritici (Black rust of wheat), Colletotrichum falcatum (Red rot of sugarcane).
Symbiotic Association: General account of Lichens and their significance.

UNIT–IV

Bryophyta: General characters; systematic position, structure, reproduction, and life cycle of Riccia (Hepaticopsida); Anthoceros (Anthocerotopsida) and Funaria (Bryopsida) excluding developmental stages.

Suggested Readings:

1. Alexopoulos, C.J., Mims, C.W., and Blackwell, M. 1996. ***Introductory Mycology.*** John Wiley and Sons, Inc. USA.
2. Dube, H.C. 1990. ***An Introduction to Fungi.*** Vikas Publishing House Pvt. Ltd., Delhi.
3. Goyal, J.P., Dhand, Neelam, and Saini, Aruna. 2011. ***Foundations of Botany.*** Trueman Book Company, Jalandhar.
4. Pandey, B.P. 2009. ***Botany for Degree Students.*** S. Chand & Co. Ltd., New Delhi.
5. Sharma, O.P. 1992. ***Textbook of Thallophytes,*** McMillan India Ltd., New Delhi.
6. Sharma, P.D. 1991. ***The Fungi.*** Rastogi & Co., Meerut.
7. Smith, G.M. 1971. ***Cryptogamic Botany. Vol. 1, Algae & Fungi,*** Tata McGraw Hill Publishing Co., New Delhi.
8. Singh, V., Pandey, P.C., and Jain, D.K. 2012. ***Textbook of Botany, Diversity of Microbes and Cryptogams.*** Rastogi Publications, Meerut & New Delhi.
9. Srivastava, H.N. 2013. ***Pradeep's Botany Vol. I (Diversity of Microbes and Cryptogams),*** Pradeep Publications, Jalandhar (India).
10. Vashishta, B.R. 1999. ***Botany for Degree Students. Algae.*** S. Chand and Company Ltd., New Delhi.
11. Vashishta, B.R. 1999. ***Botany for Degree Students. Fungi.*** S. Chand and Company Ltd., New Delhi.
12. Puri, P. 1980. ***Bryophyta.*** Atma Ram & Sons, Delhi.
13. Vashishta, B.R. 1999. ***Botany for Degree Students. Bryophyta.*** S. Chand and Company Ltd., New Delhi.

Practical:

1. Study of morphology of various genera included in algae and fungi.
2. Study of Crustose, Foliose, and Fruticose types of Lichen thalli.
3. Histopathological study of White rust of crucifers, Black rust of wheat, and Red rot of sugarcane.
4. Riccia: Study of morphology of thallus, Study of L.S. Mature sporogonium (through permanent slides) and Preparation of permanent stained slide of V.S. Riccia thallus.
5. Anthoceros: Study of morphology of thallus, Study of T.S. Thallus passing through antheridia, T.S. Thallus passing through archegonia, L.S. Mature sporogonium (through permanent slides) and Preparation of permanent stained slide of V.S. thallus.
6. Funaria: Study of morphology, Study of L.S. Male receptacle, L.S. Female receptacle, L.S. Capsule (through permanent slides) and Preparation of permanent stained slide of T.S. of stem.

Semester-II

Course Title: PLANT DIVERSITY-II Course Code : BOTDSC-2

Objective:
The basic objective of this paper is to make the students aware of the diversity in various life forms of the plant kingdom. A sequential study ranging from Pteridophytes - the first vascular land plants to gymnosperms, would enable students to have a broad perspective of evolutionary trends in the plant kingdom.

Max. Marks: 100 **Course Credits: 3 + 1 = 4**

1. **Theory: 75** **Written Exam: 60** **Internal Assessment: 15**
2. **Practical: 25** **Written Exam: 20** **Internal Assessment: 5**

Periods per week:

1. **Theory:** Four of 45 minutes duration (Three Hours)
2. **Practical:** One of two hours duration (Two Hours)

Instructions for Paper Setters:
Students will attempt five questions out of a total of nine. Question No. 1 will be COMPULSORY and comprise 8 short-answer type questions of one and a half marks each and will cover the whole of the syllabus. Units 'I', 'II', 'III' and 'IV' will have two questions each and one question is to be attempted out of each unit.

Instruction Aids: Teachers can use lecture, lecture-cum-demonstration, project, assignments, problem-solving, inductive and deductive methods, and can use L.C.D., L.E.D., multimedia projector in addition to black/whiteboard.

UNIT–I

Pteridophyta: General characters; systematic position, structure, reproduction, and life cycle of Rhynia (Psilophytopsida) and Selaginella (Lycopsida) excluding developmental stages, heterospory, and seed habit.

UNIT-II

Systematic position, structure, reproduction, and life cycle of Equisetum (Sphenopsida) and Pteris (Pteropsida) excluding developmental stages.

UNIT–III

Gymnosperms: General characters and economic importance of gymnosperms; systematic position, structure, reproduction, and life cycle of Cycas.

UNIT-IV

Systematic position, structure, reproduction, and life cycle of Pinus and Ephedra.

Suggested Readings:

1. Goyal, J.P., Dhand, Neelam, and Saini, Aruna 2011. ***Foundations of Botany.*** Trueman Book Company, Jalandhar.
2. Pandey, B.P. 2009. ***Botany for Degree Students.*** S. Chand & Co. Ltd., New Delhi.
3. Sharma, O.P. 1990. ***Textbook of Pteridophyta,*** McGraw Hill Publishing Co., New Delhi.
4. Singh, V., Pandey, P.C., and Jain, D.K. 2012. ***Textbook of Botany, Diversity of Microbes and Cryptogams.*** Rastogi Publications, Meerut & New Delhi.
5. Smith, G.M. 1971. ***Cryptogamic Botany. Vol. II, Bryophytes and Pteridophytes,*** Tata McGraw Hill Publishing Co., New Delhi.
6. Srivastava, H.N. 2013. ***Pradeep's Botany Vol. I (Diversity of Microbes and Cryptogams),*** Pradeep Publications, Jalandhar (India).
7. Bhatnagar, S.P., and Moitra, A. (1996) ***Gymnosperms.*** New Age International (P) Ltd. Publishers, New Delhi, India.
8. Chopra, G.L. ***Textbook of Gymnosperms.*** S. Nagin, Delhi, 1976.
9. Sporne, K.R. ***The Morphology of Gymnosperms,*** Hutchinson & Co. (Publishers) Ltd., London, 1965.

Practical:

1. *Selaginella:* Study of morphology, study of L.S. sporangiferous spike (through permanent slide), and preparation of permanent slide of T.S. stem.
2. *Equisetum:* Study of morphology, study of L.S. strobilus and T.S. strobilus (through permanent slide), and preparation of permanent slide of T.S. aerial stem passing through internode.
3. *Pteris:* Study of morphology, study of mature prothallus (through permanent slide), and preparation of permanent slide of T.S. petiole.
4. *Cycas:* Morphology (coralloid root, scale and foliage leaves, male cone, megasporophyll). Preparation of permanent slide of Cycas (T.S. leaflet).
5. *Pinus:* Morphology (dwarf shoot, male and female cone, and seed), preparation of permanent slide of Pinus (T.S. needle).
6. *Ephedra:* Morphology (male and female cone), preparation of permanent slide of Ephedra (T.S. stem).

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Chemistry

Semester-1
CHMDSC1: Chemistry

Total Marks: 100 (75(T) + 25(P))
Total Credits: 04 (03(T) + 01(P))

Theory

Time: 3 Hrs.
Max Marks: 75 (60 + 15)
45 Hrs. (3 Hrs. /week)

OBJECTIVE OF THE COURSE

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (GENERAL) (Semester System) in the subject of Chemistry has been upgraded as per the provision of the UGC module and the demand of the academic environment. The course contents have been revised from time to time as per the suggestions of the teachers of Chemistry working in Panjab University, Chandigarh, and affiliated colleges. The syllabus contents are duly arranged unit-wise, and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

INORGANIC CHEMISTRY (15 Hrs.)

Atomic Structure

Idea of de-Broglie matter waves, Heisenberg uncertainty principle, atomic orbitals, Schrodinger wave equation, the significance of ψ and ψ^2, quantum numbers, radial and angular wave functions (excluding mathematical relations), probability distribution curves, shapes of *s, p, d* orbitals. Aufbau and Pauli exclusion principles, Hund's multiplicity rule. Electronic configurations of the elements and ions.

Periodic Properties

Position of elements in the periodic table; effective nuclear charge and its calculations, atomic and ionic radii, ionization energy, electron affinity, and electronegativity– definition, methods of determination or evaluation, trends in the periodic table, and applications in predicting and explaining chemical behavior.

ORGANIC CHEMISTRY (15 Hrs.)

Structure and Bonding

Hybridization, Bond lengths and bond angles, Bond Energy, Localized and Delocalized chemical bond, Van der Waal's interaction, Resonance and Resonance effect, Hyperconjugation, Inductive and Field effect, Electromeric effect, Hydrogen Bonding.

Reactive Intermediates

Curved arrow notation, Drawing electron movements with arrows, half-headed and double-headed arrows, homolytic and heterolytic bond breaking, Types of Reagents – Electrophiles and Nucleophiles, Types of Organic Reactions.

Reactive intermediates- Carbocations, Carbanions, Free Radicals, Carbenes, Arynes, and Nitrenes (with examples). Assigning Formal charges on intermediates and other ionic species.

Geometrical and Conformational Isomerism

Geometrical isomerism

Cause and conditions for geometrical isomerism. Nomenclature of geometrical isomers- cis & trans, E and Z system. Determination of the configuration of geometrical isomers.

Conformational Isomerism

Representation of conformations-Sawhorse and Newman formulae. Conformational analysis of ethane, propane, n-butane, cyclohexane. Equatorial and Axial bonds. Conformations of Mono and Disubstituted derivatives of cyclohexanes.

PHYSICAL CHEMISTRY (15 Hrs.)

Basic Concepts of Mathematics:

Logarithmic relations, differentiation, and integration of functions like e^x, x^n, sin x, log x. Terms of mean and median, precision, and accuracy in chemical analysis.

Chemical Kinetics-I

Chemical kinetics and its scope, rate of a reaction, factors influencing the rate of a reaction - concentration, temperature, pressure, solvent, light, catalyst. Concentration dependence of rates, mathematical characteristics of simple chemical reactions – zero order, first order, second order, pseudo order, half-life, and mean life. Determination of the order of reaction – differential method, method of integration, method of half-life period, and isolation method.

Radioactive decay as a first-order phenomenon.

Chemical Kinetics-II

Theories of Chemical Kinetics: Effect of temperature on rate of reaction, Arrhenius equation, concept of activation energy. Simple collision theory based on hard sphere model, transition state theory (equilibrium hypothesis). Expression for the rate constant based on equilibrium constant and thermodynamic aspects.

Catalysis and general characteristics of catalytic reactions, Homogeneous catalysis, acid-base catalysis, and enzyme catalysis including their mechanisms, Michaelis-Menten equation for enzyme catalysis and its mechanism.

Instructions for Examiner:

- Question paper will consist of 7 questions in total.
- The first question will be compulsory and will carry short questions of 12 marks.
- There will be 2 questions from each unit, carrying 16 marks each.
- Candidates will be asked to attempt one question from each unit.

Suggested Books:

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., ***Basic Inorganic Chemistry***; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., ***Concise Inorganic Chemistry***; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., ***Inorganic Chemistry***; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., ***Concepts and Models of Inorganic Chemistry***; 3rd edition, Pubs: John Wiley and Sons Inc., 1994.
5. Porterfield, W.W., Wesky, A., ***Inorganic Chemistry***; Pubs: Addison-Wesley Publishing Company, 1984.
6. Miessler, G.L., Tarr, D.A., ***Inorganic Chemistry***; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Jolly, W.L., ***Modern Inorganic Chemistry***; 2nd edition, Pubs: Tata McGraw-Hill Publishing Company Limited, 1991.
8. Purcell, K.F., Kotz, J.C., ***Inorganic Chemistry***; Pubs: W.B. Saunders Company, 1977.
9. Puri, B.R., Sharma, L.R., Kalia, K.K., ***Principles of Inorganic Chemistry***; 30th edition, Pubs: Milestones Publisher, 2006-07.
10. Atkins, P., Paula, J. de, ***Atkins Physical Chemistry***; 8th edition, Pubs: Oxford University Press, 2008.
11. Puri, B.R., Sharma, L.R., Pathania, M.S., ***Principles of Physical Chemistry***; 43rd edition, Pubs: Vishal Publishing Co., 2008.
12. Barrow, G.M., ***Physical Chemistry***; 6th edition, Pubs: McGraw Hill Inc., 1996.
13. Rao, C.N.R., ***University General Chemistry***; Pubs: Macmillan India, 1985.
14. Berry, R.S., Rice, S.A., Ross, J., ***Physical Chemistry***; 2nd edition, Pubs: Oxford University Press, 2000.
15. Albert, R.A., Silbey, R.J., ***Physical Chemistry***; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
16. Dogra, S.K., Dogra, S., ***Physical Chemistry Through Problems***; Pubs: Wiley Eastern Limited, 1991.
17. Levine, I.N., ***Physical Chemistry***; 5th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
18. Moore, W.J., ***Basic Physical Chemistry***; Pubs: Prentice Hall of India Pvt. Ltd, 1983.
19. Metz, C.R., ***Theory and Problems of Physical Chemistry***; Schaum's Outline Series, 2nd edition, Pubs: McGraw-Hill Book Company, 1989.

Semester-1 (Major)
CHMDSC1: LABORATORY PRACTICALS-I

Max. Marks: 25
Total Credits: 01

QUANTITATIVE ANALYSIS:

Volumetric titrations involving acid-base, $KMnO_4$
There are 2 experiments: one involving acid-base titrations, one involving $KMnO_4$, and one involving $K_2Cr_2O_7$.

1. Determination of the strength of Na_2CO_3 solution by titrating it against a standard solution of HCl.
2. Determination of the molarity of $KMnO_4$ solution by titrating it against a standard solution of oxalic acid.

ORGANIC CHEMISTRY AND GREEN CHEMISTRY PRACTICALS

Crystallization and determination of melting points

Concept of induction of crystallization:

1. Phthalic acid from hot water (using fluted filter paper and stemless funnel).

PHYSICAL CHEMISTRY

1. **Viscosity**
   To determine the viscosity of Brine Solution (20%), n-Butyl alcohol, cyclohexane

Semester-2

CHMDSC2: Chemistry

Total Marks: 100 (75(T) + 25(P))
Total Credits: 04 (03(T) + 01(P))

Theory

Time: 3 Hrs.
Max Marks: 75 (60 + 15)
45 Hrs. (3 Hrs. /week)

OBJECTIVE OF THE COURSE

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (GENERAL) (Semester System) in the subject of Chemistry has been upgraded as per the provision of the UGC module and the demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of Chemistry working in Panjab University, Chandigarh, and affiliated colleges. The syllabus contents are duly arranged unit-wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

INORGANIC CHEMISTRY (15 Hrs.)

Chemical Bonding – I
Covalent Bond – Valence bond theory and its limitations, directional characteristics of covalent bond, various types of hybridization, and shapes of simple inorganic molecules and covalent ions: BeF_2, BF_3, CH_4, PF_5, SF_6, IF_7, $SnCl_2$, XeF_4. Valence shell electron pair repulsion (VSEPR) theory to H_3O^+, SF_4, ClF_3, ICl_4^-, and H_2O. MO theory, homonuclear (elements and ions of 1st and 2nd row) and heteronuclear (CN, CO^+, CO, CN^-) diatomic molecules. Percentage ionic character from dipole moment and electronegativity.

Chemical Bonding – II
Ionic Solids – Concept of close packing, ionic structures (NaCl type, and Zinc blende), radius ratio rule and coordination number, limitation of radius ratio rule, lattice defects, semiconductors. Lattice energy and Born-Haber cycle, solvation energy and solubility of ionic solids, polarizing power and polarizability of ions, Fajan's rule. Metallic bond: free electron, valence bond, and band theories. Weak Interactions – Hydrogen bonding, Van der Waals forces.

ORGANIC CHEMISTRY (15 Hrs.)

Alkanes and Cycloalkanes
Structure, IUPAC Nomenclature, and Isomerism. Methods of Preparation (Sabatier and Senderen's

reaction, from alkyl halide, Wurtz reaction, Corey-House synthesis, Kolbe's method, decarboxylation), Physical and Chemical Properties.
Mechanism of Halogenation of Alkanes – Relative reactivities of halogens, alkanes, and different classes of hydrogen atoms; Orientation, Reactivity, and Selectivity.

Stereochemistry of Organic Compounds – Optical Isomerism
Concept of Isomerism. Types of isomerism. Tetrahedral concept of carbon, Polarimetry, specific rotation. Optical isomerism – Elements of Symmetry, Molecular Chirality, Enantiomerism, stereogenic center, optical activity, chiral and achiral molecules with two stereogenic centers, Diastereoisomers, Erythro and Threo isomers, Meso compounds. Representation of Stereoisomers – Saw-horse, Newman, Flying edge, and Fischer Projection. Number of stereoisomers in chiral compounds, Racemisation, Internal and External compensation, Resolution of racemic mixture, Enantioselective synthesis, Walden inversion.
Relative and absolute configuration, sequence rules, D & L, and R & S system of nomenclature (with examples).

PHYSICAL CHEMISTRY (15 Hrs.)

Gaseous States
Gas laws, Ideal and real gases, postulates of kinetic theory of gases, deviation from ideal behavior, Van der Waals equation of state. Critical Phenomena: PV isotherms of real gases, the isotherms of Van der Waals equation, relationship between critical constants and Van der Waals constants, the law of corresponding states, reduced equation of state.
Molecular Velocities: Root mean square, average, and most probable velocities. Collision number, mean free path, and collision diameter. Liquefaction of gases (based on Joule-Thomson effect).

Thermodynamics I
Definition of Thermodynamic Terms: System, surroundings, etc. Types of systems, intensive and extensive properties. State and path functions and their differentials. Thermodynamic process. Concept of heat and work.
First Law of Thermodynamics: Statement, definition of internal energy and enthalpy, Heat capacity, heat capacities at constant volume and pressure and their relationship. Joule's Law – Joule-Thomson coefficient and inversion temperature. Calculations of w, q, dU & dH for the expansion of ideal gases under isothermal and adiabatic conditions for a reversible process.

Instructions for the Examiner:

- The question paper will consist of 7 questions in total.
- The first question will be compulsory and will carry short questions of 12 marks.
- There will be 2 questions from each unit, carrying 16 marks each.
- The candidate will be asked to attempt one question from each unit.

Books Suggested:

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., *Basic Inorganic Chemistry*; 2nd edition, Pubs: John Wiley and Sons, 1995.
2. Lee, J.D., *Concise Inorganic Chemistry*; 4th edition, Pubs: Chapman & Hall Ltd., 1991.
3. Shriver, D.E., Atkins, P.W., *Inorganic Chemistry*; 4th edition, Pubs: Oxford University Press, 2006.
4. Douglas, B., Medaniel, D., Atenander, J., *Concepts and Models of Inorganic Chemistry*; 3rd edition, Pubs: John Wiley and Sons Inc., 1994.
5. Porterfield, W.W., Wesky, A., *Inorganic Chemistry*; Pubs: Addison-Wesley Publishing Company, 1984.
6. Miessler, G.L., Tarr, D.A., *Inorganic Chemistry*; 3rd edition, Pubs: Pearson Education Inc., 2004.
7. Jolly, W.L., *Modern Inorganic Chemistry*; 2nd edition, Pubs: Tata McGraw-Hill Publishing Company Limited, 1991.
8. Purcell, K.F., Kotz, J.C., *Inorganic Chemistry*; Pubs: W.B. Saunders Company, 1977.
9. Puri, B.R., Sharma, L.R., Kalia, K.K., *Principles of Inorganic Chemistry*; 30th edition, Pubs: Milestones Publisher, 2006-07.
10. Atkins, P., Paula, J. de, *Atkins Physical Chemistry*; 8th edition, Pubs: Oxford University Press, 2008.
11. Puri, B.R., Sharma, L.R., Pathania, M.S., *Principles of Physical Chemistry*; 43rd edition, Pubs: Vishal Publishing Co., 2008.
12. Barrow, G.M., *Physical Chemistry*; 6th edition, Pubs: McGraw Hill Inc., 1996.
13. Rao, C.N.R., *University General Chemistry*; Pubs: Macmillan India, 1985.
14. Berry, R.S., Rice, S.A., Ross, J., *Physical Chemistry*; 2nd edition, Pubs: Oxford University Press, 2000.
15. Alberty, R.A., Silbey, R.J., *Physical Chemistry*; 1st edition, Pubs: John Wiley & Sons Inc., 1992.
16. Dogra, S.K., Dogra, S., *Physical Chemistry Through Problems*; Pubs: Wiley Eastern Limited, 1991.
17. Levine, I.N., *Physical Chemistry*; 5th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
18. Moore, W.J., *Basic Physical Chemistry*; Pubs: Prentice Hall of India Pvt. Ltd., 1983.
19. Metz, C.R., *Theory and Problems of Physical Chemistry*; Schaum's Outline Series, 2nd edition, Pubs: McGraw-Hill Book Company, 1989.

Semester-2

CHMDSC2 :LABORATORY PRACTICALS-II

Max. Marks: 25

Total Credits: 01

QUANTITATIVE ANALYSIS:

Volumetric titrations involving acid-base, $KMnO_4$ and $K_2Cr_2O_7$.

There are three experiments-one involving $K_2Cr_2O_7$.

1. Standarise the given $K_2Cr_2O_7$ solution by titrating it against a standard solution of Mohr's Salt.

ORGANIC CHEMISTRY AND GREEN CHEMISTRYPRACTICALS

Crystallization and determination of melting points

Concept of induction of crystallization

1. Acetanilide from boiling water.
1. Benzoic acid from water

PHYSICAL CHEMISTRY

a. Surface Tension

To determine the surface tension of Brine Solution (20%), n-Butyl alcohol, cyclohexane

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Computer Science and Applications

Title : Computer Fundamentals and PC Software
Paper Code: CSADSC1

L	T	P	Cr
2	-	4	4

Time Duration: 3 Hrs.

Theory External Marks: 45
Theory Internal Marks : 05
Number of Th. Hrs. : 30
Practical Ext. Marks : 50
Number of Pr. Hrs. : 60

Objectives: The objective of this course is to familiarize students with the fundamentals of computers and packages commonly used in application software.

Course Outcome :

On successful completion of the course, students will be able to:

i. Understand the fundamentals of personal computers, application usages etc.
ii. Get the knowledge of peripherals of personal computer and booting process.
iii. Use word processing packages for writing emails, letters and chapters etc.
iv. Handle statistical data using spreadsheets packages.
v. Design presentations for business analyses, industrial growth etc.

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT - I

Introduction to Computers: Characteristics of Computers, Functional Units of Computer System: CPU, registers, system bus, main memory unit, cache memory; Inside a computer: SMPS, Motherboard, Ports and Interfaces, expansion cards, ribbon cables, memory chips, processors

Devices: Input and output devices, Keyboard, Mouse, Joystick, Scanner, OCR, OMR, Bar Code Reader, Web Camera, Monitor, Printer and Plotter; Memory: Primary, Secondary/Auxiliary memory, RAM and ROM and their types, Cache memory, Units of measurement of storage; Storage devices, Hard disks, Optical disks, Memory storage devices: Flash Drive, Memory card; Types of software: System and Application software

Data Representation: Bit, Byte, Binary, Decimal, Hexadecimal, and Octal Systems. Conversions between different number systems; Character Codes: Introduction to ASCII, EBCDIC and Unicode character sets

Understanding Windows Operating System: Anatomy of windows, Taskbar Settings, managing folders and files, Searching Files and folders, Customizing Windows, Recycle Bin, Control Panel.

UNIT - II

Introduction to Word Processing: Opening, Creating, Saving, Printing and closing Documents, Using the Interface (Menu Toolbars), Editing Text (Copy, Delete, Move), Finding and Replacing Text, Spell Check, Autocorrect; Auto-text, Character formatting, Paragraph formatting, Page formatting.
Document Enhancement: Inserting Pictures, Tables, inserting special characters, Adding Borders and shading, Adding Headers and Footers, numbering pages, setting up Multiple columns, Adjusting Margins and Hyphenating Documents, changing document views, Adding comments to a document, Tracking changes to a document.
Using styles and templates: Applying, modifying and creating new (custom) styles; Creating a table of contents, creating indexes and bibliographies
Mail Merge and Macros: Creating Master Document and Data Source, Merging and printing Documents; Creating and using Macros, Creating fill-in forms.

UNIT - III

Introduction to Spreadsheets: Worksheet overview, Row, Column, Cells, Ribbon, Tabs, Groups, Name Box and Formula Bar; Creating, Opening, Saving and printing worksheet; Speeding up data entry: using fill tool, fill series, defining fill series; working with Formulae, Data formatting (number formatting, date formatting),Working with Ranges, Absolute, relative and Mixed addressing; Viewing a spreadsheet: freezing rows and columns, splitting screen; creating, sorting and filtering Data Base; Creating and running Macros, Assigning shortcuts to Macros
Using Functions: Statistical, financial, Mathematical, string, date and time.
Working with sheets: inserting new sheet, moving, deleting and renaming sheets
Creating Charts: Understanding Chart types, Creating Embedded Charts, formatting Charts: Changing Chart types, adding Titles, Legends and Gridlines, Printing Charts

UNIT – IV

Presentation Software: Creating, saving, and printing presentations; selecting design templates, Changing Background and Layout, Inserting tables and images, animations and transitions
Databases: Introduction to database, Creating database using Wizard or from scratch, creating tables using wizard, entering data, using design view, saving, inserting, editing, Changing properties of fields, setting primary key.

Suggested Readings:

1. Sinha P.K. & Sinha Priti, Computer Fundamentals, BPB Publications
2. Prof. Satish Jain, Shashi Singh, M. Geetha, Windows 10 with MS office 2016, BPB Publications
3. Mansfield Ron,Working in MS Office, Tata McGraw Hill.

Title : Practical Based on Paper – CSA01
(Computer Fundamentals and PC Software)
Paper Code: CSADSC2

Time Duration: 3 Hrs. Number of Hours: 60

Note : Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

Title : Programming in C
Paper Code :CSADSC3

L	T	P	Cr
2	-	4	4

Time Duration: 3 Hrs.

Theory External Marks: 45
Theory Internal Marks : 05
Number of Th. Hrs. : 30
Practical Ext. Marks : 50
Number of Pr. Hrs. : 60

Objectives: The objective of this course is to familiarize students with programming language "C" and its use in Problem solving.

Course Outcome :

On successful completion of the course, students will be able to:

i. Understand the programming concepts.
ii. Learn the C programming in practical usage.
iii. Solve programming problems using C programming language.
iv. Gain the in depth knowledge of strings, arrays, functions and file handling etc.
v. Develop applications based on programming concepts.

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT - I

Programming Process: Steps in developing of a program, Algorithm development, Flowchart, Testing and Debugging.

Fundamentals of C Languages: History of C, Character Set, Identifiers and Keywords, Constants, Types of C Constants, Rules for Constructing Integer, Real and character Constants, Variables, Data Types, rules for constructing variables.

Operators and Expressions: C Instructions, Arithmetic operators, Relational operators, Logical operators, Assignment Operators, Type Conversion in Assignments, Hierarchy of Operations, Standard and Formatted Statements, Structure of a C program, Compilation and Execution.

UNIT - II

Decision Control Structure: Decision making with IF-statement, IF-Else and Nested IF-Else, The else if Clause.

Loop Control Structure: While and do-while, for loop and Nested for loop, Case Control Structure: Decision using switch, The goto statement.

Functions: Library functions and user defined functions, Global and Local variables, Function Declaration, Calling and definition of function, Methods of parameter passing to functions, recursion.

UNIT - III

Arrays: Introduction, Array declaration, accessing values in an array, initializing values in an array, Single- and Two-Dimensional Arrays, initializing a 2-Dimensional Array, Memory Map of a 2-Dimensional Array, Arrays of characters, Insertion and deletion operations, Searching the elements in an array, Using matrices in arrays.

Pointers: Pointer declaration, Address operator "&", Indirection operator "*", Array of Pointers.

UNIT - IV

Storage Classes in C :Auto, Extern ,Register and Static with their Scope, Storage & Lifetime.

String Manipulation in C: Declaring and Initializing string variables, Reading and writing strings, String Handling functions (strlen(), strcpy(), strcmp(), strcat()).

Structures and Unions: Definition. Advantages of Structure, Declaration of structures, Structure Initialization, Accessing Structure members, Arrays of Structures, Union Definition, Difference between Structure and Union.

Suggested Readings:

1. Yashavant P. Kanetkar : Let us C, BPB Publications, New Delhi.
2. Salaria, R.S. : Test Your Skills in C, Salaria Publications, New Delhi.
3. C. Balaguruswami : Programming with C Language, Tata McGraw Hill, New Delhi.
4. Byron S. Gottfried : Programming in C, McGraw Hills Publishers, New York.
5. M.T. Somashekara : Programming in C, Prentice Hall of India.

Title : Practical Based on Paper – CSA03
(Programming in C)
Paper Code: CSADSC4

Time Duration: 3 Hrs. Number of Hours: 60

Note : Paper will be set at the time of examination. Due weight-age may be given to the practical note-book and Assignments in evaluation.

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ENVIRONMENT CONSERVATION

SEMESTER-I

ENVIRONMENTAL CONSERVATION: NATURAL RESOURCE MANAGEMENT

Course Code: ENCDSC1 **Credits= 4 (T=3 + P=1)**
Course Title: Environmental Conservation: Natural Resources management
Total Hours = 60 (Theory=45+Practical=15)
Total Marks = 100 (Theory=75+Practical=25)
Internal Assessment =20%

Course Outcomes

1. *The students will be able to understand about natural resources, their types and realize the importance of resources.*
2. *The students will be able to acquire deep knowledge about land and forest resources.*
3. *The students will be able to describe energy resources.*
4. *The students will be able to analyse Indian as well as global statusof natural resources.*
5. *The students will be able to study conservation methods of natural resources.*

THEORY

Unit-I (15 Hrs)
Natural and Energy Resources:
1.1 Natural resources-Classification and types of resources
1.2 Renewable and non-renewable resources
1.3 Concept of conventional and non-conventional sources of energy
1.4 Fossil fuels- coal, crude oils and natural gas
1.5 Solar energy
1.6 Wind energy
1.7 Biomass energy
1.8 Geothermal energy

Unit-II (15 Hrs)
Forest Resources

2.1 Importance of forest;
2.2 Productive functions, protective and ameliorative functions,
2.3 Non-timber forest products
2.4 Major forest types in India
2.5 Conservation of forest resources
2.6 Global status of forest conservation efforts

Unit-III (15 Hrs)
Land Resources
3.1 Land Resources: Global land use patterns
3.2 Soil degradation – causes; Soil erosion, Desertification

3.3 Wasteland Reclamation. Land degradation
3.4 Land cover, land use changes; land degradation and its control
3.5 Water logging and salinity
3.6 Organic farming

PRACTICALS (15 Hrs)

1. Prepare a report on promotion of green energy in India
2. To do a survey of water consumption and conservation practises of a selected area.
3. To visit and make a list of forest and non-forest products in the market.
4. Identify the sources of energy used in your area
5. Visit to a local area to document environmental assets river/ forest/grassland/hill/mountain
6. To determine soil quality parameters using portable soil analysis kit

SUGGESTED READINGS:

1. Rogoff, M., Screve, F. 2019. Waste-to-Energy, Elsevier, Amsterdam
2. Singh, M.P, Rallan,B.R. and Vivek Kumar. 2012. Natural Resources Management. Emkay Publishing House.
3. Odum, E.P. and Barrett, G.W. 2008. Fundamentals of Ecology (5thEdition). Thomson Brooks Australia and Affiliated to East West Press Pvt. Ltd., New Delhi.
4. Kanagasabai, T. 2010. Environmental Studies. PHI Learning Private Limited, New Delhi. 5. Sharma, P.D. 2018. Ecology and Environment. Rastogi Publication, Meerut.
6. Joseph, K. and Nagendran, R. 2004. Essentials of Environmental Studies. Pearson Education, New Delhi.

SEMESTER-II

ENVIRONMENTAL CONSERVATION: ENVIRONMENTAL POLLUTION AND MANAGEMENT

Course Code: ENCDSC2 **Credits= 4 (T=3 + P=1)**
CourseTitle: Environmental Conservation: Environmental Pollution and Management
Total Hours = 60 (Theory = 45 + Practical = 15)
Total Marks = 100 (Theory = 75 + Practical = 25)
Internal Assessment = 20%

Course Outcomes

1. *The students will be able to understand about causes, sources and effects of environmental pollution.*
2. *Thestudentswillbeableto learn about atmospheric structure and processes*
3. *ThestudentswillacquireknowledgeaboutGlobal Water Balance and Hydrological cycle.*
4. *Thestudentswill get practical knowledge of water quality parameters and soil parameters.*

THEORY

Unit-I (15 Hrs)

Hydrosphere and Water Resources

1.1 Composition of Hydrosphere, Global Water Balance
1.2 Hydrological Cycle; Major Sources of Water, Surface and subsurface water sources
1.3 Water Anomalies including Floods and Droughts
1.4 Overutilization of Water and Water Scarcity
1.5 Conflicts Over Water – National and International
1.6 Case Studies; Satluj Yamuna Link Canal Issue, Kaveri Issue

Unit-II (15 Hrs)

Water Pollution and Control Measures

2.1 Water Pollution: Concept, Causes and Sources
2.2 Water Pollutants and categories
2.3 Effect of water pollution on plants , animals and humans
2.4 Basics of Water Testing , Potable water
2.5 Control Measures; Waste Water Treatment
2.6 Root Zone technology

Unit-III(15 Hrs)

Air Pollution and Control Measures

3.1 Composition and Structure of Atmosphere
3.2 Air Pollution: Concept and Sources; Noise Pollution
3.3 Air Pollutants and Categories
3.4 Effects of Air Pollution
3.5 Major Air Pollution Issues: Global Warming, Acid rain, Ozone Depletion
3.7 Control Measures; Air Pollution Control Devices

PRACTICALS (15 Hrs)

1. Draw the pie diagram showing the global water balance.
2. To study the structure of the atmosphere with the help of a label diagram
3. To determine water quality parameters (pH, EC and TDS) using analysis kit.
4. To study air quality parameter like SPM using Air Quality Monitor
5. To visit, collect data and prepare report regarding air and water pollution of a site.

SUGGESTED READINGS:

1. Joseph, K. and Nagendran, R.(2004). Essentials of Environmental Studies. Pearson Education, New Delhi.
2. Cunningham, W.P. and Cunningham, M.A. (2002*). Environmental Science: Inquiry andApplications.* AGlobalConcern.TataMcGraw-HillPublishingCompany,NewDelhi.
3. De. AK, (2003). Environmental Chemistry. New Age International, New Delhi
4. Khitoliya, RK (2012) Textbook on Environmental Pollution. S Chand Publishing, New Delhi
5. American Public Health Association, Standard Methods for Examination of Water and Wastewater, APHA, New York, 2003.
6. Sengupta, B., John, S. & Abraham, T. Forth Estate, A Handbook on Air, Water and Land Pollution. IDMA Foundation for Sustainable Development and BR Publications Corporation, Delhi. 2006.
7. Aggarwal, A. K., Noise Pollution. APH Publications, New Delhi, 2008.
8. Dara, S.S., A Textbook of Environmental Chemistry. 2nd Edition. S. Chand & Company Ltd., Ram Nagar, New Delhi, 2010.
9. Hill, M. K., Understanding Environmental Pollution, Cambridge Univ. Press, 2008.
10. Aggarwal, A. K., Water Pollution, APH Publications, New Delhi, 2008

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Foreign Trade: Policy and Procedure

SUBJECT: OBJECTIVE

1. To familiarize the students with the basic principles of foreign trade and the environment in which foreign trade takes place.

2. To familiarize the students with the position of India's foreign trade, Import and export policies and various export promotion measures adopted by the Government.

3. To familiarize the students with the nature and scope of International Marketing and also the four Ps of international Marketing.

4. To familiarize the students with the various methods and procedure of foreign trade, financing, foreign exchange, rates, costing and pricing for exports and the various institutions involved in export finance.

5. To make the students aware of the shipping and insurance practices and procedures which constitute the essential services for the operation of foreign trade.

6. To familiarize the students with the basic documents involved in foreign trade, processing of an export order, customs clearance of export and import cargo and negotiation of documents.

Job Potential

- Self- employment can start an export business either singly or in partnership with fellow students.
- Can take up export documentation work for others.
- Can take up employment in exporting firms, firms, banks, Insurance companies or with freight forwarders.

SEMESTER – I

FOREIGN TRADE: POLICY AND PROCEDURE (FTPP)
PAPER – I, BASICS OF FOREIGN TRADE

TOPOLOGY OF CODE	MAJOR / MINOR
COURSE DURATION / LTP	L = 60 HOURS 04 HOURS A WEEK FOR 15 WEEKS – 1ST SEMESTER
CREDITS	4
MAXIMUM MARKS	100 MARKS = THEORY 90 MARKS + INTERNAL ASSESMENT 10 MARKS
TIME	3 HOURS

Instructions for Paper-setter and candidates:
The maximum marks for the paper will be 100. The question paper will be of 90 marks and internal assessment will be of 10 marks. Time allowed will be 3 hours.

The paper-setter must put a note in the question paper in this regard. There shall be 5 questions in all.
The first question is compulsory comprising of 06 short answer type questions spread over the whole syllabus. The candidates are required to attempt 05 questions. Each question shall be of two marks (5x2= 10).
Rest of the paper shall contain two units. Each unit will have 04 questions. The candidates will be asked to attempt two questions from each unit. Each question will carry 20 marks (20x4=80)

FOREIGN TRADE : POLICY AND PROCEDURE
Contents for Semester Courses

SEMESTER 1

Paper l -BASICS OF FOREIGN TRADE
Objectives: To familiarize the students with the basic principles of foreign trade and the environment in which foreign trade takes place.

Unit-I

- Foreign Trade :Definition, objectives and importance
- Theories of international trade
- Balance of trade and Balance of Payments
- Objectives of Trade Policy and role of foreign trade in economic growth
- International economic institutions - GATT, UNCTAD, IMF and World Bank.

Unit-II

- Instruments of trade Policy tariffs, quantitative restrictions, exchange etc.
- Control and exchange rate adjustments.
- Trade Blocs and Regional Economic Cooperation
- Cultural Aspects in International Business

Suggested Readings
International Economics by P.T. Ellsworth International Economics by C.P. Kindelberger

B.A./ B. Sc. (General)

Semester 2

Paper II - India's Foreign Trade 100

Note: Practical Training after 1st year (2nd Semester) and Viva-Voce. 25

Semester 3

Paper III - Elements of Export Marketing 100

Semester 4

Paper IV - Foreign Trade Financing and Procedure 100

Note: Practical Training 2nd year (4th Semester) and Viva-Voce. 25

Semester 5

Paper V - Shipping and Insurances Practices and Procedures 100

Semester 6

Paper VI - Foreign Trade Documentation and Procedures 100

Examination: Paper will be divided into three Sections; Section A will consist of Six Short answer type questions and students would be asked to attempt any four, section B and Section C will have four essay type questions from Unit I and Unit II each. Students will be asked to attempt at least two from each section.

Duration : 3 hours-Theory

SEMESTER-II

FOREIGN TRADE: POLICY & PROCEDURE (FTPP)

Paper-II INDIA's FOREIGN TRADE

NOTE- PRACTICAL TRAINING IN SECOND SEMESTER AND VIVA-VOCE

TOPOLOGY OF CODE	MAJOR / MINOR
COURSE DURATION / LTP (LECTURES, TUTORIALS, PRACTICALS)	L = 60 HOURS, 04 HOURS A WEEK FOR 15 WEEKS – 2nd SEMESTER
CREDITS	4
MAXIMUM MARKS	100 MARKS = THEORY 65 MARKS + INTERNAL ASSESMENT 10 MARKS 25 MARKS= PRACTICAL TRAINING & VIVA-VOCE
TIME 3 HOURS	3 HOURS FOR THEORY

FOREIGN TRADE: POLICY & PROCEDURE

Content for Semester Course
Semester II

Paper-2 INDIA's FOREIGN TRADE

Unit-I

- Analysis of India's Foreign -Trade Growth trends, composition and direction.
- India 's balance of Payments including invisibles
- Assessment of Prospects -Products and markets
- India's Trade agreements

Unit-II

- Salient features of India's export-import policy
- Institutional set-up for export promotion
- Export Assistance measures, free trade zones and 100%EOUs
- State trading in India
- Sources and analysis of foreign trade Statistics

Suggested Reading

Annual Reports of the Ministry of Commerce, Annual Economics: Surveys, Import and Export Policy

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Industrial Chemistry

Semester 1

Total Marks: 100(75(T)+25(P))
Total credits: 04 (03(T)+01(P))

ICHDSC- Basic Industrial Principles and Laboratory Operations, Data Analysis using A.I.

Max. Marks: 75
Total Credits: 03

Theory

OBJECTIVEOFTHECOURSE

To teach the fundamental concepts of Industrial Chemistry and their applications. The syllabus pertaining to B.Sc. (Semester system) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I

Dimensions and Units - Definitions of the Basic Units - (Mass, Length, Time, Temperature, Volume, Pressure), Derived units, Conversion between units, Significant figures.

Basic Chemical Calculations - Atomic weight, Molecular weight, Equivalent weight, Mole, Mole fraction, Mole percent, Mass fraction of solution, Weight percentage.

UNIT-II

Chemical Concentration– Definition, Percentage (mass percent, volume percent, Mass-volume percent), Parts per millions (ppm), Strength of solution, Molarity, Molality, Normality, Formality.

Preparation of solutions - Standard solutions, Primary standards, Secondary standards.

Stoichiometric Analysis – Stoichiometry, Limiting reactant, Excess reactant, Yield.

UNIT-III

Common Laboratory Apparatus and Calibration – Theory of Analytical Balance, Operation and theory of the balance, Construction details, Components, Sources of Errors and Error Analysis, Maintenance Guidelines.

Fundamentals of Laboratory Apparatus - Volumetric flasks, burettes, pipettes, weighing bottles, specific gravity bottle, different types of funnels, chromatographic columns, chromatographic jars, desiccators, drying ovens, filter crucibles, rubber policeman, use of volumetric glass ware.

UNIT-IV

pH meter - components of pH meter, use of pH Meter, Calibration and maintenance of pH meter, application of data.

Errors in Chemical Analysis - Types of errors, Accuracy and Precision.

Data analysis and Chemical drawing and visualization tool- Excel, Origin software, ChemDraw, ChemSketch.

Safety with chemicals and waste.

Books suggested

1. Higson, S. P.J. (2003), Analytical Chemistry, Oxford University Press.
2. Skoog, D.A.; West, D.M. (2003), Fundamentals of Analytical Chemistry, Brooks/Cole.
3. Christian, G.D.(2004), Analytical Chemistry, 6th Edition, John Wiley & Sons, New York.
4. Fifield,F.W.; Kealey, D. (2000), Principles and Practice of Analytical Chemistry, Wiley.
5. Dean J. A. (1997), Analytical Chemistry Handbook, McGraw Hill.
6. Day. R. A.; Underwood, A. L. (1991), Quantitative Analysis, Prentice Hall of India.
7. Gordus, A. A. (1985), Schaum's Outline of Analytical Chemistry, Tala McGraw-Hill.
8. Harris, D. C. (2007), Exploring Chemical Analysis, W.H. Freeman and Co.

Industrial Chemistry
ICHDSC (LABORATORYPRACTICALS-I)

Max. Marks: 25
Total Credits: 01

Chemistry Lab: Basic Industrial Laboratory Operations

1. Use and calibration of volumetric equipment and glassware.
2. Preparation of standard solutions of NaOH, $KMnO_4$, H_2SO_4, HCl, Na_2CO_3, Oxalic acid. Introduction to primary and secondary standards.
3. Calibration of equipment.
4. Use of pH meter: determination of pH of given solutions.

Books suggested

1. Seamus P.J. Higson: Analytical Chemistry.
2. Douglas A. Skoog and Donald M..West: Fundamentals of Analytical Chemistry.
3. Adion A. Gordus: Schaum's Outline of Analytical Chemistry, Tata McGraw-Hill.
4. Gary D. Christian : Analytical Chemistry .
5. Freifelder and Kealy: Analytical Chemistry .
6. Daniel C Harris: Exploring Chemical Analysis.
7. Daniel C Harris: Quantitative Chemical Analysis.

Industrial Chemistry
Semester 1
DS-Minor-IND CHM 1A
Total Marks: 100(75(T)+25(P))
Total credits: 04 (03(T)+01(P))

DS-Minor-IND CHM 1A-Theory-Basic Industrial Principles and Laboratory Operations

Max. Marks: 75
Total Credits: 03

OBJECTIVEOFTHECOURSE

To teach the fundamental concepts of Industrial Chemistry and their applications. The syllabus pertaining to B.Sc. (Semester system) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I

Dimensions and Units - Definitions of the Basic Units - (Mass, Length, Time, Temperature, Volume, Pressure), Derived units, Conversion between units, Significant figures.
Basic Chemical Calculations - Atomic weight, Molecular weight, Equivalent weight, Mole, Mole fraction, Mole percent, Mass fraction of solution, Weight percentage.

UNIT-II

Chemical Concentration– Definition, Percentage (mass percent, volume percent, Mass-volume percent), Parts per millions (ppm), Strength of solution, Molarity, Molality, Normality, Formality.
Preparation of solutions - Standard solutions, Primary standards, Secondary standards.
Stoichiometric Analysis – Stoichiometry, Limiting reactant, Excess reactant, Yield.

UNIT-III

Common Laboratory Apparatus and Calibration – Theory of Analytical Balance, Operation and theory of the balance, Construction details, Components, Sources of Errors and Error Analysis, Maintenance Guidelines.
Fundamentals of Laboratory Apparatus - Volumetric flasks, burettes, pipettes, weighing bottles, specific gravity bottle, different types of funnels, chromatographic columns, chromatographic jars, desiccators, drying ovens, filter crucibles, rubber policeman, use of volumetric glass ware.

UNIT-IV

pH meter - components of pH meter, use of pH Meter, Calibration and maintenance of pH meter, application of data.
Errors in Chemical Analysis - Types of errors, Accuracy and Precision.
Safety with chemicals and waste.

Books suggested

1. Higson, S. P.J. (2003),Analytical Chemistry, Oxford University Press.
2. Skoog, D.A.; West, D.M. (2003), Fundamentals of Analytical Chemistry, Brooks/Cole.
3. Christian, G.D.(2004), Analytical Chemistry, 6th Edition, John Wiley & Sons, New York.
4. Fifield,F.W.; Kealey, D. (2000), Principles and Practice of Analytical Chemistry, Wiley.
5. Dean J. A. (1997), Analytical Chemistry Handbook, McGraw Hill.
6. Day. R. A.; Underwood, A. L. (1991), Quantitative Analysis, Prentice Hall of India.
7. Gordus, A. A. (1985), Schaum's Outline of Analytical Chemistry, Tala McGraw-Hill.
8. Harris, D. C. (2007), Exploring Chemical Analysis, W.H. Freeman and Co.

Industrial Chemistry

DS-Minor-IND CHM 1A (LABORATORYPRACTICALS-I)

Max. Marks: 25

Total Credits: 01

Chemistry Lab: Basic Industrial Laboratory Operations

1. Use and calibration of volumetric equipment and glassware.
2. Preparation of standard solutions of NaOH, $KMnO_4$, H_2SO_4, HCl, Na_2CO_3, Oxalic acid. Introduction to primary and secondary standards.
3. Calibration of equipment.
4. Use of pH meter: determination of pH of given solutions.

Books suggested

1. Seamus P.J. Higson: Analytical Chemistry.
2. Douglas A. Skoog and Donald M..West: Fundamentals of Analytical Chemistry.
3. Adion A. Gordus: Schaum’s Outline of Analytical Chemistry, Tata McGraw-Hill.
4. Gary D. Christian : Analytical Chemistry .
5. Freifelder and Kealy: Analytical Chemistry .
6. Daniel C Harris: Exploring Chemical Analysis.
7. Daniel C Harris: Quantitative Chemical Analysis.

Industrial Chemistry
Semester 2
Major/Minor

Total Marks: 100(75(T)+25(P))
Total credits: 04 (03(T)+01(P))

ICHDSC2- Applied Industrial Chemistry

Max. Marks: 75
Total Credits: 03

Theory

OBJECTIVEOFTHECOURSE

To teach the fundamental concepts of Industrial Chemistry and their applications. The syllabus pertaining to B.Sc. (Semester system) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I

Oils & Fats: Introduction, Extraction of oils from vegetable oils, refining of oils and fats, hydrogenation of oils.

Soaps and Detergents: Introduction, Raw materials, Manufacture of soap, Classification of detergents, finishing of detergents.

UNIT-II

Pulp & paper: Introduction, Raw Materials, types of pulp, Manufacture of paper.

Sugar: Introduction; Sugar extraction, defacation, sulphitation, carbonation, concentration, crystallization, drying, refining; Uses of molasses and bagasse.

UNIT-III

Cement: Cement-Types of cement, Constituents of cement, Manufacture of Portland cement.

Ceramics: Introduction, Properties of ceramics, Classification of refractories, Important steps involved in the manufacture of refractories.

UNIT-IV

Cosmetics and Perfumes - Introduction to cosmetics and perfumes, preparation and uses of the following: Hair dye, hair spray. shampoo, face powder, lipsticks. talcum powder, nail enamel.

Essential oils and their importance in cosmetic industries with reference to Eugenol. Geraniol, Sandalwood oil, Eucalyptus, Rose oil.

Books suggested

1. Shreev, R.N. & Brink, J.A. : Chemical Process Industries, 5th Edition, McGraw Hill, 1987.
2. Austine, G.T. : Shreeves Chemicals Process Industries, 5th Edition, McGraw Hill, 1984.
3. Dryden, C.E., Rao M.G. & Silting, M.: Outlines of Chemical Technology, 3rd Edition, affiliated East West Press Pvt. Ltd., N.Delhi, 2008.

4. Pandey, G.N. : Chemical Technology, Volume-II, Lion Press, Kanpur.
5. Donnet J. B., Bansal R. C. : Carbon Fibres, Marcel Dekker Inc.
6. Donnet J. B., Bansal R. C., Wang M. J.
7. Bansal R. C., Donnet J. B., Stoeckli F. : Carbon Black, Marcel Dekker Inc. : Active carbon, Marcel Dekker Inc.

Industrial Chemistry Semester 2 (Major)
(LABORATORYPRACTICALS-II)

Max. Marks: 25
Total Credits: 01

Chemistry Lab: Basic Industrial Laboratory Operations

1. Oils & Fats: Determination of Acid value, Iodine value, Saponification value.
2. Carbohydrates: Reducing and non reducing sugars by
   (i) Fehlings method (ii)Pavy's method.
3. Cement: Loss of ignition, silica, insolubles, ash content.
4. Preparation of nail polish and nail polish remover.

Books suggested

1. Vermani, O. P.; Narula, A. K. (2004), Industrial Chemistry, Galgotia Publications Pvt. Ltd., New Delhi.
2. Bhatia, S. C. (2004), Chemical Process Industries, Vol. I & II, CBS Publishers, New Delhi.
3. Barel, A.O.; Paye, M.; Maibach, H.I.(2014), Handbook of Cosmetic Science and Technology, CRC Press.
4. Gupta, P.K.; Gupta, S.K.(2011),Pharmaceutics and Cosmetics, Pragati Prakashan
5. Butler, H. (2000),Poucher's Perfumes, Cosmetic and Soap, Springer

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Mathematics

Semester-I

MATDSC1:ALGEBRA &TRIGONOMETRY

Credits=4 (L =3, P=1)

One credit would mean either a lecture (L) of 1 Hour/ Week or a practical (P) of 3 hours/week. So, 4 credits would correspond to 3 hours of lectures and 3 hours of practicals per week. Total contact hours in a semester of 15 weeks will be 90 hours. (45 hrs. lectures + 45 hrs. practicals)

Exam Time: 3 hours

Max.Marks:100

(External: 90 + Internal assessment : 10)

Note: 1.The syllabus has been split into Four Units. The examiner will set in total Nine questions including a compulsory question. The compulsory question (QuestionNo.1)will contain6 parts covering the entire syllabus. Two questions will be set from each Unit.

2. A student will be required to attempt five questions in all, selecting one questions from each Unit and the compulsory question. Each question will carry equal marks.

Objective: The main goal of this course is to deliver basics of Binomial Theorem, Properties of Binomial coefficients . The course introduces the concept of De Moivre's theorem and its applications, eigen values and eigen vectors of a matrix. The objective is to understand nature of solution of a linear system of equations.

Learning Outcomes: The students will be able to determine the sum of infinite Binomial Series, the rank of a matrix, eigen values, eigen vectors and solve linear system of equations using matrices.

UNIT-I

Principle of Mathematical Induction, Recall of Binomial Theorem for positive index, General and Middle terms in Binomial expansions, Properties of Binomial coefficients, Binomial Theorem for any index, Summation of infinite Binomial series.

UNIT-II

Solution of Trigonometric Equations, Sine, Cosine and Projection formulae for triangles,

De Moivre's theorem, applications of De Moivre's theorem including primitive nth root of unity. Expansions of sin nθ , cos nθ , $\sin^n\theta$, $\cos^n\theta$ ($n \in \mathbb{N}$). The exponential, logarithmic, direct and inverse circular and hyperbolic functions of a complex variable.

UNIT –III

Recall of determinant of a matix, Properties of determinants, Hermitian and skew-hermitian matrices, linear dependence of row and column vectors, row rank, column rank and rank of a matrix and their equivalence.

UNIT-IV

Theorems on consistency of a system of linear equations (both homogeneous and non-homogeneous). Eigen-values, eigen-vectors and characteristic equation of a matrix, Cayley-Hamilton theorem and its use in finding inverse of a matrix. Diagonalization.

References:

1. K. B. Datta: Matrix and Linear Algebra, Prentice Hall of India Pvt. Ltd., New Delhi, 2004.
2. S. R. Knight and H. S. Hall: Higher Algebra, H. M. Publications, Seventh Edition, 1994.
3. R.S. Verma and K.S. Shukla: Text Book on Trigonometry, PothishalaPvt. Ltd., Allahabad, Tenth Edition, 1999.
4. Shanti Narayan and P.K. Mittal: A Text Book of Matrices, S. Chand & Co., New Delhi, Revised Edition, 2007.

Semester-II
MATDSC2: CALCULUS-I

Credits=4 (L =3, P=1)

One credit would mean either a lecture (L) of 1 Hour/ Week or a practical (P) of 3 hours/week. So, 4 credits would correspond to 3 hours of lectures and 3 hours of practicals per week. Total contact hours in a semester of 15 weeks will be 90 hours. (45 hrs. lectures + 45 hrs. practicals)

Exam Time: 3 hours

Max.Marks:100
(External: 90 + Internal assessment : 10)

Note: 1.The syllabus has been split into Four Units. The examiner will set in total Nine questions including a compulsory question. The compulsory question (QuestionNo.1)will contain6 parts covering the entire syllabus. Two questions will be set from each Unit.
2. A student will be required to attempt five questions in all, selecting one questions from each Unit and the compulsory question. Each question will carry equal marks.

Objective: The main goal of this course is to deliver basics of Limits, Continuity and Differentiability of functions. Besides this differentiation of hyperbolic and inverse hyperbolic with their applications will also be studied. The course will cover the Mean Value Theorems and their applications.

Learning Outcomes: After completing this course the students will be able to compute limits and continuity of various functions. The course will empower the students to compute derivatives of Hyperbolic functions. The students will be able to compute tangents and normals.

UNIT-I

Order properties of real numbers, bounds, l.u.b. and g.l.b. order completeness property of real numbers, Archimedean property of real numbers.
Limits: ε - δ definition of the limit of a function, basic properties of limits, infinite limits,

UNIT-II

Continuous functions, types of discontinuities, continuity of composite functions, continuity of $|f(x)|$, sign of a function in a neighborhood of a point of continuity, intermediate value theorem, maximum and minimum value theorem.

UNIT-III

Hyperbolic, inverse hyperbolic functions of a real variable and their derivatives, successive differentiations, Leibnitz's theorem.
indeterminate forms.
Applications of Derivatives :Tangents and normals, Differentials and Approximations, Errors.

UNIT- IV

Mean value theorems:Rolle's Theorem, Lagrange's mean value theorem, Cauchy's mean value theorem, their geometric interpretation and applications, Taylor's theorem, Maclaurin's theorem with various form of remainders and their applications.

References

1. Murray R. Spiegel: Schaum's outline series Theory and Problems of Advanced Calculus, Schaum Publishing Co. New York, Second Edition, 2002.
2. P. K. Jain and S. K. Kaushik: An Introduction to Real Analysis, S. Chand & Co., New Delhi, First Edition, 2000.
3. G. Prasad: Differential Calculus, PothishalaPvt Ltd., Allahabad, Revised Edition, 2016.
4. G. B. Thomas and R. L. Finney: Calculus and Analytic Geometry, Pearson Publication, Ninth Edition, 2010.
5. Shanti Narayan and P. K. Mittal: Differential Calculus, S. Chand & Co., New Delhi, Fifteenth Edition, 1942.
6. R. G. Bartle, D. R. Sherbert: Introduction to Real Analysis, Wiley India Pvt. Ltd., Fourth Edition, 2021.

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MICROBIOLOGY

SEMESTER-I
MICDSC1: Basic Microbiology

Total marks: 75
Theory: 67
Internal assessment: 08
Time: 3 hours
Credits: 3

Note: The question paper will consist of four Units (I-IV).There will be nine questions, out of which five questions haveto be attempted. Question I will span the complete syllabus and will be compulsory. Rest of the eight questions will be from different Units of the syllabus. There will be two questions from each Unit and one is to be attempted from each Unit.Each question should be sub-divided into 2-4 sub parts.

Objective And Learning outcome: Aim of the course is to introduce the students with microbial world and to make the students to learn the morphology, microscopy, staining. It will also enrich the students' knowledge on the development of microbiology as the discipline.

UNIT–I

NO. OFHOURS: 15

HISTORY OF MICROBIOLOGY
Development of microbiology as a discipline, Spontaneous generation vs. biogenesis. Contributions of Antoni van Leeuwenhoek, Louis Pasteur, Robert Koch, Joseph Lister, Alexander Fleming. Role of microorganisms in fermentation, Germ theory of disease.

MICROSCOPY
Microscopic examination of micro-organism, bright field microscopy, dark field microscopy, phase contrast microscopy and electron microscopy.

STAINING
Nature of stains, principles, mechanism, methods and types of staining: simple, gram staining, capsule staining and acid fast staining.

UNIT–II

NO.OF HOURS: 15

MORPHOLOGY AND FINE STRUCTURE OF BACTERIA
Cell size shape and arrangement, Organization of cell wall, cell membrane, flagella and capsules, fimbriae and pili, mesosomes, nucleoid chromosome and plasmids in bacteria.

BACTERIOLOGICALTECHNIQUES
Pure culture isolation: Culture media, Streaking, serial dilution and planting methods; cultivation, maintenance and preservation/stocking of pure cultures; cultivation of anaerobic bacteria.

UNIT-III

NO. OF HOURS:15

MICROBIALGROWTH:
Phases of growth, generation time, growth rate, bacterial growth curve, mathematical nature and expression, diauxic growth, chemostat and turbidostatic. Factors affecting growth in microorganisms. Methods for measurement of growth in microorganisms, formation of endospores in bacteria.

UNIT-IV

NO.OFHOURS:15

CONTROL OF MICROORGANISMS

Physical and chemical methods of sterilization, disinfection, antimicrobial agents, sulfa drugs, antibiotics: broad-spectrum antibiotics, beta-lactam antibiotics.

NORMAL MICROBIAL FLORA

Germfree and gnotobiotic life, distribution and occurrence of the normal flora : skin, eye, respiratory tract, mouth, intestinal tract, genitourinary tract.

MIN :101 Basic Microbiology (PRACTICALS)
SEMESTER-I

Total marks: 25
Practical: 22
Internal assessment: 03
Time: 3 hours
Credits :1

List of the Practical:

1. Preparation of different media: Synthetic Media, Complex media (Nutrient Agar, McConkey agar).
2. Simple staining.
3. Negative staining.
4. Capsule staining.
5. Isolation of pure cultures of bacteria by streaking method.
6. Estimation of CFU count by spread plate method/pour plate method.
7. Demonstration of Motility by hanging drop method.

SUGGESTEDREADINGS

1. Black JG. (2008). Microbiology: Principles and Explorations. 7th edition. Prentice Hall
2. Madigan MT, and Martinko JM. (2014). Brock Biology of Micro-organisms. 14thedition. ParkerJ.Prentice Hall International, Inc.
4. Pelczar Jr MJ, Chan ECS, and Krieg NR. (2004). Microbiology. 5th edition Tata McGraw Hill.
5. Srivastava S and Srivastava PS. (2003). Understanding Bacteria. Kluwer Academic Publishers, Dordrecht.
6. Stanier RY,Ingraham JL, Wheelis ML and Painter PR. (2005). General Microbiology. 5th edition McMillan.
7. Tortora GJ, Funke BR, and Case CL. (2008). Microbiology:An Introduction. 9th edition Pearson Education.
8. Willey JM,Sherwood LM, and Woolverton CJ. (2013). Prescott's Microbiology. 9th edition. McGraw Hill Higher Education.
9. Cappucino J and Sherman N.(2010).Microbiology:A LaboratoryManual9thedition.Pearson Publication

Semester-II

MICDSC2: Microbial Physiology

Total marks: 75
Theory: 67
Internal Assessment: 08
TIME: 3 hours
Credits:3

Note: Question paper will have four Units(I-IV). Examiner will set a total of nine questions comprising two questions from each unit and one compulsory question of short answer type covering the whole syllabus. The students will attempt one question from each unit and the compulsory question. All questions may carry equal marks, unless specified.

Objectives: The course gives an overview of structure,biological importance of carbohydrates, lipids, biological enzymes, proteins and their properties.Also to understand DNA and RNA ,types of RNA and their biological functions.

UNIT –I **15 HOURS**

Carbohydrates:

Monosaccharides: Classification based on aldehyde and ketone groups; structure of Ribose, Deoxyribose, Glucose, Galactose and Fructose.

Disaccharides: Glycosidic bond, structure of lactose and sucrose.

Polysaccharides : Structure and biological role of starch, glycogen and cellulose.

Lipids :

1. Simple lipids – Fats and oils, waxes. 2. Compound lipids – Phospholipid, Glycolipids 3. Derived lipids – Cholesterol

Enzymes:

1. Definition, 2. Structure- Concept of apoenzyme, coenzyme, cofactor 9 and active site. 3. Types- Extracellular, Intracellular, Constitutive and Inducible. 4. Features of enzyme - substrate reaction.

UNIT –II **15 HOURS**

Nucleic Acids :

DNA – structure and composition (Watson and Crick Model) 2. RNA – Types (m-RNA, t-RNA, r-RNA), structure and functions.

Proteins :

Structural levels of proteins: primary, secondary, tertiary and quaternary domain structure,protein classification and function.

UNIT –III **15 HOURS**

Enzymes :

Chemical and physical properties of enzymes. Classification and nomenclature of Enzymes. Mechanisms of enzymes action. Inhibition, control and regulation of enzyme activity. Factors affecting enzyme activity. Enzyme kinetics: Michaelis-Menten equation.

UNIT–IV **15 HOURS**

ATP generation

Substrate level phosphorylation, Oxidative phosphorylation, Bacterial Photophosphorylation – Cyclic and Non- cyclic ,Catabolism of glucose – EMP, Tricarboxylic Acid Cycle, Pentose Phosphate Pathway, Entner Doudoroff Pathway.

MICDSC2: MICROBIAL PHYSIOLOGY(PRACTICALS)
SEMESTER-II

MAX. MARKS: 25
PRACTICAL: 22
INTERNAL ASSESSMENT: 03
TIME: 3 HRS.
CREDITS:1

List of Experiments:

1. Qualitative Analysis of Carbohydrates.
2. Colorimetric estimation of proteins by Biuret / Lowry method.
3. Separation of mixtures by paper / thin layer chromatography.
4. Effectoftemperature/pH / Salt concentration on bacterial growth.
5. Demonstration of electrophoretic technique.
6. Study and plot the growth curve of E. coli by turbidometric and Standard Plate Count methods.

SUGGESTED READINGS:

1. Berg JM, Tymoczko JL and Stryer L (2011) Biochemistry, W.H.Freeman and Company Caldwell, D.R. (1995). Microbial Physiology and Metabolism, W.C. Brown Publications,Iowa, USA.
2. Sashidhara Rao, B. and Deshpande, V. (2007). Experimental Biochemistry: A student Companion. I.K. International Pvt. Ltd.
3. Tymoczko JL, Berg JM and Stryer L (2012) Biochemistry: A short course, 2nd ed.,
4. W.H.Freeman 6. Voet,D. and Voet J.G (2004) Biochemistry 3rd edition, John Wiley and Sons
5. Fisher, M. (2001). Lehninger principles of biochemistry, ; by David L. Nelson and Michael M. Cox. The Chemical Educator, 6, 69-70

Office Management and Secretarial Practice

1	**Course Title**	**OFFICE MANAGEMENT AND METHODS**
2	**Course Code**	OMSDCS1
3	**Typology of Course**	DISCIPLINE SPECIFIC
4	**Course Duration/ LTP** (Lectures, Tutorials, Practical)	**30 hours** of **lectures** for **Theory + 60 hours of Practical** (1 credit = 15 hours of **theory learning** based on 1 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits Theory + Practical = 2+2 credits
6	**Max. Marks and Time**	100 Marks (45 TH + 5 IA + 50 P)
7	**Course Objectives**	1. To enable the students to understand the working and procedures of the Modern Office 2. To develop the necessary skills required for the smooth functioning of the office and to perform various office Operations efficiently
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1: Understand office Management, Procedures and Practices being followed in a Modern Office CO2: Understand office automation, space management, workplace environment CO3: understand the machines and equipment being used in Modern Office for automation of Office functioning
9	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Practicals and Group Discussions etc.
10	**Instructions for paper setters and candidates**	**Note :** The question paper of each subject covering the entire course shall be divided into three sections **Section A (14 marks)** This section will have 10 short answer questions from the entire syllabus. Students are required to attempt 7 questions from this section. Each question will carry 2 marks; the total weightage being 15 marks. **Section B (16 marks)** This section will consist of essay type from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks. **Section C (15 marks)** This section will consist of essay type from Unit II of the syllabus. The candidate will be required to attempt three questions out of five questions. Each question will carry 5 marks; the total weightage being 15 marks

11 Course Content

UNIT – I

Introduction of Office and Office Management : Meaning, Importance and Functions of Modern Office; Centralization and Decentralization of Office Services - advantages and disadvantages; The Changing Office Scene - The Concept of Paperless, Automated and Virtual Office ; Introduction, Functions and Elements of Office Management; Functions of Office Manager, Essential Qualification, Experience and Supervisory Skills required for Office Manager.

Office Layout : Introduction and Objectives of Layout, Principles of Office Layout, Factors to be considered in Physical Layout; Concept, Advantages and Drawbacks of Open and Private Offices, New Trends in Office Layout; Ergonomics of the Office : Office Lighting - Natural and Artificial Light, Ventilation and Temperature Control, Noise - Internal and External, Office Furniture, Cleanliness and Sanitary Arrangements, Security of Employees

UNIT – II

Office Automation : Meaning, advantages and disadvantages of Mechanization, Factors in selection of Office Machines; Modern Office machines (Their Uses & Operations) : Computer and its Peripherals like Scanner and Printers (Laser, Inkjet, Impact), Plotter, Multimedia Devices (Digital Camera, Projector), Photocopier, Spiral Binding Machine, Lamination System, Telephone, Intercom, Facsimile machine, PBX, EPABX, Internet.

Office Forms : Introduction, Importance and Advantages of Office Forms, Types of Forms, Principles of Form Designing, Computerization of Office Forms (with special reference to e-Forms)

Office Stationery: Methods of Purchasing Stationery, Purchase Procedure, Control on Consumption of Office Stationery. Management of Stationery in Stock Register.

Practical work:

- The students should visit different modern offices (minimum two) in order to understand the concept of an office. After the visit, a Project Report indicating different departments of office, functions of office, functions of Office Manager etc. to be prepared by each student.
- The students are required to study different Organizational Charts online and prepare at least one sample of the same.
- During the visit mentioned above, the students should observe the office layout, Environment aspects of the office, physical amenities provided to the staff etc. The changing scenario in the office being visited has to be observed by the students and their views presented in the Project Report.
- The students should create their own E-mail Accounts and send letters to their friends and keep a record of the same in their project files (Minimum five mails).

- The students should practice filling up of different types of forms – offline and online. (At least five different type of forms)
- The student should practice creating e-Forms, adding questions, images, sections, importing questions, collecting and downloading responses and changing settings

12 Suggested Readings

- Chhabra, T.N., Modern Business Organisation, New Delhi, DhanpatRai& Sons.
- Duggal, Balraj, Office Management and Commercial Correspondence, KitabMahal, New Delhi.
- P.K. Ghosh, "Office Management", Sultan Chand & Sons. New Delhi
- R.K. Chopra, Office Management, Himalaya Publishing House

1	**Course Title**	**TYPOGRAPHY AND COMPUTER APPLICATION-I**
2	**Course Code**	OMSDCS1
3	**Typology of Course**	DISCIPLINE SPECIFIC
4	**Course Duration/ LTP** (Lectures, Tutorials, Practical)	**30 hours** of **lectures** for **Theory + 60 hours of Practical** (1 credit = 15 hours of **theory learning** based on 1 hrs./week) (1 credit = 30 hours of **practical** based on 2 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits Theory + Practical = 2+2 credits
6	**Max. Marks and Time**	100 Marks (45 TH + 5 IA + 50 P)
7	**Course Objectives**	• To familiarize the students with the principles of Typography. • To develop practical understanding among the students through QWERTY keyboard operations through touch system of typewriting. • To enable the students to understand the fundamentals of Computer System, its hardware & software components • To have working knowledge of different parts of Computer • To enhance the understanding of usefulness of Information Technology tools for business operations.
8	**Course Outcomes**	**On successful completion of the course, the Students will be able to** CO1: Understand the fundamentals of computer. CO2: Understand the Internet Technologies CO3: Learn the concepts of word-processing (MS-Office) transactions & articulate the Royalty agreements.
9	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Practicals and Group Discussions etc.
10	**Instructions for paper setters and candidates**	**Note :** The question paper of each subject covering the entire course shall be divided into three sections **Section A (14 marks)** This section will have 10 short answer questions from the entire syllabus. Students are required to attempt 7 questions from this section. Each question will carry 2 marks; the total weightage being 15 marks. **Section B (16 marks)** This section will consist of essay type from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 8 marks; the total weightage being 16 marks.

	Section C (15 marks) This section will consist of essay type from Unit II of the syllabus. The candidate will be required to attempt three questions out of five questions. Each question will carry 5 marks; the total weightage being 15 marks
11 Course Content	**UNIT – I** Introduction to Information Technology : Computer System, Characteristics of Computer, Capabilities and limitations of Computers, Generations of computers, Types of Computers; Components of Computer System, Block Diagram of Computer, Input and Output Devices; Memory : Primary Storage and Secondary Storage; Types of Computer Software : System, Application and Utility Software ; Operating Systems : Types and Functions Keywords: URL, IP address, Hyperlinks; Web pages, Home page, web sites, WWW, Dial-up, Broadband, Dedicated, ISP, Browser, DSL, DNS, Gateways, Chat rooms, Downloading and Uploading, Wifi. Internet : Introduction, Host and Terminals, Types of Internet Connections, Uses of Internet, Search Engines and their Types, Browsing Information from the Web using popular Internet Search Engines; E-mail : Opening accounts; Creating, Reading, Printing, Deleting, Replying, Forwarding e-mail with or without attachment Networking : Need, Advantages, Disadvantages and Types of Computer Network; Computer Virus: Working of Viruses, Types of Viruses, Virus Detection, Prevention and Cure, antivirus Packages **UNIT – II** Sitting Posture - Chair & Table Height Guideline & Placement of Monitor & Keyboard; Methods of Keyboard Operation - Sight Method & Touch Method, Meaning and uses of Touch Method; The student is required to achieve proficiency in e-typewriting with touch method of typewriting Method of speed calculation; The minimum accurate speed to be attained is 30 words per minute; Meaning of Word Processor, Need and Uses of Word Processing, Advantages and Limitations of Word Processing; Software used for Word Processing, Why MS-Word and which version? Starting Word: MS-Word interface, opening a blank document, hiding and showing toolbars, templates. Working in Word: selecting text, editing text, finding and replacing text, formatting text, checking and correcting spellings, Justification and Alignment, Bullets and Numbering, Tabs, Paragraph formatting, Indent, Page Formatting, Header and Footer & Word Count; Working with a Document: Page Setup of a document, viewing a document, switching between documents, saving a document, print preview, printing a document; Finishing Touch to a document: Inserting date and time, Special effects such as Bold, Scripts, etc., Inserting a Picture, Inserting Scanned and Digital Camera Images, Creating & Using Auto Text Entries, Adding Tables to the Documents, Adding New Row in a Table, Drawing Tables Freehand, Adding Borders (to Table as well as Page); Using Mail Merge

		Practical work: • E-typewriting : A paragraph of 300 words to be typed in 10 minutes • A question on MS-Word comprising of simple formatting of passage/letter e.g. bold, italics etc. • General mail merge for the letter so created to send it to multiple recipients at the same time • Working with Microsoft Internet Explorer: Opening a web page, opening multiple browser windows, opening multiple tabs in a single browser windows and their management, working offline, deleting temporary files, exploring Internet Options
12	**Suggested Readings**	• Absolute Beginner's Guide to Computer Basics, Michael Miller. • Fundamental of Computers, AkashSaxena, Kratika Gupta. • Fundamentals of Information Technology, Alexis and Mathew. • Computer Fundamentals, P.K. Sinha. • Principles of Typewriting, D.P. Bhatia and S.S. Sangal. • Microsoft Word 2010 Step by Step(Microsoft) by Joyce Cox and Joan Lambert. • MS Word 2000 Thumb Rules and Details, Snigdha Banerjee. • Word 2010 All-in-One For Dummies, Doug Lowe and Ryan C. Williams

Physics

Semester 1

PHYDSC1: Mechanics

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 3 Hrs per week
Total Credits: 3 (Theory)+1(Practical)

OBJECTIVE OF THE COURSE

To teach the fundamental concepts of Mechanics and their applications. The syllabus pertaining to B.Sc. in the subject of Physics has been upgraded as per provision of the NEP-2020 and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Physics working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Cartesian and spherical polar co-ordinate systems, area, volume, displacement, velocity, and acceleration in these systems, solid angle.Centre of mass, linear momentum, angular momentum, torque, potential energy, and kinetic energy of a system of particles. Relationship of conservation laws of linear momentum, angular momentum and energy, symmetries of space and time.

Various forces in nature, relative strengths and spatial dependence, Motion under force obeying inverse square law, equivalent one-body problem. Motion under central forces, equation of motion under central force, equation of orbit and turning points, Kepler's Laws. Elastic collision in lab and C.M. systems, relationships of velocities, angles, and kinetic energies in these two systems.

UNIT-II

Rigid Body motion; Rotational motion, principal moments and Axes, Euler equations. Galilean transformations and Invariance, Non-Inertial frames. Fictitious forces in a rotating frame of reference, Centrifugal and Coriolis forces due to rotation of earth.Concept of stationery universal frame of reference and ether, Michelson-Morley experiment, and its results.

Postulates of special theory of relativity, Lorentz transformations, Kinematical consequences of Lorentz transformations – length contraction and time dilation, Twin paradox, Transformation of velocities,Relativistic Doppler effect.Variation of mass with velocity, mass-energy equivalence, rest mass in an inelastic collision, relativistic momentum & energy, their transformation.

Books Suggested:

Essential Readings:

1. *Mechanics*, Berkeley, Vol. I, C. Kittle.
2. Mechanics, H.S. Hans & S.P. Puri.

Further Readings:

1. *An Introduction to Machines,* Daniel Kleppner& Robert J. Kolenkow (TMH).
2. *Introduction of Classical Mechanics,* R.G. Takwale & P.S.Puranik(TMH,2000).
3. *Mechanics & Relativity (3rd Edition)*, Vidwan Singh Soni (PHI Learning, New Delhi, 2013)
4. *Basic Concepts of Relativity,* R.H. Good (East-West Press, New Delhi, 1974).

SEM 1:PHYSICS PRACTICALS

Max. Marks: 25 (20+5 internal assessment)
Total Credit: 1
Teaching hours: 2 Hrs per week

The activities given in the section "Analysis of Experimental Data" are compulsory for all the students in the First semester.

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions.

(Candidates are expected to state their results with limits of error).	09
(ii) One exercise based on experiment or Computer Programming	04
(iii) Viva-Voce	04
(iv) Record (Practical file)	03
(v) Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.

3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.

4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.

5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.

6. Number of candidates in a group for practical examination should not exceed 12.

7. In a single group, no experiment to be allotted to more than three examinees in the group.

Analysis of Experimental Data (Compulsory for all students in first semester):

Objectives :

(i) Knowledge of propagation of errors.
(ii) Knowledge of significant figures, Determination of standard deviation and probable error and their use in interpretation of the experimental result.
(iii) Familiarity with the method of least square fitting of experimental data to a curve.

LIST OF EXPERIMENTS:
MECHANICS

I. Measurements :

Objectives :

(i) Measurements of time, length, thickness and curvature, pressure, humidity
(ii) Concepts of least count, horizontal, vertical and angular alignments

Activities :

(i) To measure internal/external diameter of a hollow cylinder using Vernier calipers
(ii) To measure thickness of wire
(iii) To measure curvature of a lens
(iv) To measure pressure using Barometer
(v) To measure humidity using dry and wet thermometer

II. Rotation :

Objectives :

(i) Study of rotational motion.
(ii) Establishing relationship between different quantities.

Activities :

(i) To study the dependence of moment of inertia on distribution of mass (by noting time periods of oscillations using objects of various geometrical shapes but of same mass).

(ii) To establish relationship between torque and angular acceleration using fly wheel.

III. One-Dimensional Collisions :

Objectives :

(i) Conservation of linear momentum and kinetic energy in elastic collisions.
(ii) Dependence of fraction of kinetic energy transferred on the masses of colliding bodies.
(iii) Idea of coefficient of restitution.

Activities :

To determine energy transfer, coefficient of restitution and verify laws of conservation of linear momentum and kinetic energy in elastic collisions using one dimensional collisions of hanging spheres.

IV. Compound Pendulum :

Objectives :

(i) Idea of equivalent simple pendulum.
(ii) Concepts of centre of suspension and oscillation.
(iii) Dependence of time period on moment of Inertia .
(iv) Radius of gyration.
(v) Determination of g.

Activities :

(i) Measure time period as a function of distance of centre of suspension (oscillation) from centre of mass, plot relevant graphs, determine radius of gyration and acceleration due to gravity.

(ii) Find the value of g by Katers' or Bar pendulum.

V. Torsion Pendulum :

Objectives :

(i) Idea of torsional vibration, dependence of time period on M.O.I. and restoring torque.

(ii) Modulus of rigidity.

Activity :

Measure time period of oscillation of a Maxwell needle and determine modulus of rigidity of the material of a given wire.

VI. Damped Oscillator :

Objectives :

(i) Study damped oscillations.

(ii) Coefficient of damping, quality factor etc.

Activities :

To measure/obtain logarithmic decrement, coefficient of damping, relaxation time, and quality factor of a damped simple pendulum.

VII. Elasticity :

Objective :

Knowledge of elastic constants and related quantities.

Activities :

(i) Study of bending of beams and determination of Young's Modulus.

(ii) Determination of Poisson's ratio for rubber/plastic.

VIII. Standing waves :

Objective :

Standing waves on a string and in air.

Activities :

(i) Melde's experiment.

(ii) Kundt's tube.

IX. Viscosity :

Objective :

Knowledge of viscosity of liquids.

Activity :

Determination of coefficient of viscosity of a given liquid by Stoke's method and study its temperature dependence.

Computer based activities : Elementary C Language programs, flowcharts and their interpretation.

1. To print out all natural even/odd numbers from a given series of natural numbers.
2. Numerical solution of equations of motion.
3. To calculate first ten prime numbers.

Texts and Reference Books :

1. *"B.Sc. Practical Physics by C.L. Arora, S.Chand& Co. (2014).*
2. *"A Laboratory Manual of Physics for Undergraduate Classes"* by D.P. Khandelwal.
3. *"Programming with C, Schaum series"* by Byron Gottfried&JitenderChhabra

Semester-II

PHYDSC2: ELECTRICITY AND MAGNETISM

Maximum marks: 100
(Theory 75 marks & Practical 25 marks)
Theory 75 marks [60 marks+15 marks (internal assessment 20%)]
Teaching hours: 3 Hrs per week
Total Credits: 3(Theory)+1(Practical)

OBJECTIVE OF THE COURSE

To teach the fundamental concepts of Electricity and Magnetism. The syllabus pertaining to B.Sc. in the subject of Physics has been upgraded as per provision of the NEP-2020 and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Physics working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

Note:

1. The theory paper will be of 75 marks including 15 marks for internal assessment and three Hour duration.
2. The question paper for the final examination will consist of three units. Unit I-II will have THREE questions each from the corresponding units of the syllabus. Unit III in the paper will have one compulsory question consisting of 8 short answer type questions covering the whole syllabus and the students have to attempt any 6 parts of the compulsory question. Each question will have a weightage of 12 marks. The candidates will attempt five questions in all, selecting two questions each from Unit I and II and the compulsory question from unit III.
3. The question paper is expected to contain problems with a weightage of 25-30%.
4. Use of non-programmable calculators will be allowed in the examination centre.
5. The Logarithmic table will be provided in the examination centre, if required.

UNIT-I

Basic ideas of Vector Calculus, Gradient, Divergence, curl in Cartesian coordinates and their useful relations, physical significance and applications, Conservative field, Laplacian in Rectangular coordinates. Stoke's theorem, Gauss's divergence theorem, Coulomb's Law for point charges and continuous distribution of charges, electric field due to dipole, line charge, charged ring, circular disc and sheet of charge, Gauss's Law and its differential form.

Work and potential difference, Potential difference as lineintegral of field, Electric potential due to dipole, Electric field as gradient of scalar potential, curl E = 0. Poissonand Laplace's equation. Polarisation of matter, atomic and molecular dipoles, induced dipole moment and atomic polarizability.Electric susceptibility and polarization vector. Relation $K=1+\chi$, Gauss's law for dielectrics. Displacement vector, Div. D = 0, Energy stored in dielectric medium.

UNIT-II

Current and current density, equation of continuity. Microscopic form of Ohm's Law ($\mathbf{J} = \sigma\mathbf{E}$) and conductivity. Failure of Ohm's Law.Invariance of charge.**E**in different frames of reference.
Behaviour of various substances in magnetic field.Definition of M and H and their relation to free and bound currents.Permeability and susceptibility and their interrelationship.
Lorentz's force. Definition of **B**. BiotSavart's Law and its application to long straight wire, circular current loop and solenoid. Ampere's Circuital law and its application.Divergence and curl of **B**.Current density— definition, its use in calculation of change in magnetic field at a current sheet. Energy stored in magnetic field, Faraday's Law of EM induction, Displacement current, Mutual inductance and reciprocity theorem. Self inductance for solenoid

Books Suggested :

Essential Readings :

1. Electricity & Magnetism, A.S. Mahajan& A.A. Rangwala (Tata McGraw Hill).
2. Fundamentals of Electricity and Magnetism by Arthur F. Kipp.
3. Electricity and Magnetism, Berkeley Physics Course, Vol. II by E.M. Purcell.
4. Introduction to Classical Electrodynamics by David Griffith, Prentice Hall.

Further Readings:

1. Electricity & Magnetism, 4th Edition, W.J. Duffin.
2. EM Waves and Radiating Systems, Edward C. Jordan and K. G. Balmain, Prentice Hall.

SEM 2: PHYSICS PRACTICALS

Max. Marks: 25 (20+5 internal assessment)
Total Credit: 1
Teaching hours: 2 Hrs per week

The activities given in the section "Analysis of Experimental Data" are compulsory for all the students in the First semester.

General Guidelines for Physics Practical Examinations:

1. The distribution of marks is as follows:

(i) One full experiment requiring the student to take some data, analyze it and draw conclusions. (Candidates are expected to state their results with limits of error).	09
(ii) One exercise based on experiment or Computer Programming	04
(iii) Viva-Voce	04
(iv) Record (Practical file)	03
(v) Internal Assessment	05

Note for Examiners:

2. The marks scored under each head must be clearly written on the answer sheet.
3. There will be one session of 3 hours duration. The paper will have two sections. In Section-A one experiment is to be allotted by the external examiner.
4. Section–B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.
5. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.
6. Number of candidates in a group for practical examination should not exceed 12.
7. In a single group, no experiment to be allotted to more than three examinees in the group.

ELECTRICITY AND MAGNETISM

I. Objective :

Measurement of resistance, voltage, current and electric energy.

Activities:

(i) To use a multimeter for measuring AC and DC voltage and resistance.
(ii) Measurement of resistance of LDR - To study inverse-square law (concept of solid angle and inverse square law) using linear LDR and light source.
(iii) Observations and measurements using an Electric energy meter. To find wattage of given bulb or heater.
(iv) To study the efficiency of an electric kettle or heater element with varying input voltage.

II. Low Resistance Measurements :

Objectives :

(i) Inadequacy of Wheatstone bridge to measure low resistances.
(ii) Acquaintance with a method of measuring low resistances.

Activity :

To determine low resistance with Carey Fosters Bridge.

III. Magnetic Field :

Objectives :

(i) Familiarity with the magnetic field produced by a solenoid.
(ii) Dependence of solenoidal field on number of turns and current.
(iii) Permeability of air.

Activities :

To study the magnetic field produced by a current carrying solenoid using a search coil and calculate permeability of air.

IV. Electromagnetic Induction :

Objective :

Verification of laws of electromagnetic induction.

Activity :

To study the induced e.m.f. as function of the velocity of the magnet.

V. Magnetism and current :

Objectives and Activities :

Force on a conductor carrying current in a magnetic field.

VI. LCR Circuits :

Objective :

Study of phase relationship between currents and voltages in ac circuits.

Activity :

Study of phase relationships using impedance triangle for LCR circuit and calculate impedance.

VII. Resonant Circuits :

Objective :

Concepts of resonance and Q-value.

Activities :

(i) Resonance in a series LCR circuits for different R-value and calculate Q-value.
(ii) Resonance in a parallel LCR circuits for different R-value and calculate Q-value.
(iii) To determine the dielectric constant of a solid by resonance method.

VIII. Capacitance :

Objectives :

(i) Measurement of capacitance, dielectric constant.
(ii) Concept of time constant and time base circuit.
(iii) Knowledge of a-c Bridges.

Activities :

(i) Capacitance by flashing and quenching of a neon lamp.
(ii) Measurement of capacitance, determination of permittivity of a medium, air and relative permittivity by De-Sauty's bridge.

IX. Self Inductance :

Objectives :

(i) Knowledge of a-c bridges.
(ii) Concept of self-inductance.

Activities :

(i) To determine L using Anderson Bridge.

Computer based activities: Elementary C language programs, flowchart and their interpretation.

1. To rearrange a list of numbers in ascending and descending orders.
2. To compile a frequency distribution and evaluate moments such as mean; standard deviation etc.
3. To evaluate sum of finite series and the area under a curve.

Texts and Reference Books :

1. *"B.Sc. Practical Physics by C.L. Arora, S.Chand & Co. (2014).*
2. *"A Laboratory Manual of Physics for Undergraduate Classes"* by D.P. Khandelwal.
3. *"Programming with C, Schaum series"* by Byron Gottfried & JitenderChhabra

Principles and Practices of Insurance

SEMESTER-I

INSDSC1:Insurance and Risk Management

Duration: 3 Hours **Max. Marks: 90**

Internal Assessment:10

Objective: This paper will explore the role of insurance in financial planning and in risk management techniques. This will serve as a base for acquiring knowledge about the different aspects of insurance business.

Unit-I

Risk Management:

Definition of Risk, Risk and Uncertainty, Types of risks, Meaning and Importance of Risk Management, Risk Assessment Process, Risk management Techniques, Challenges in Risk Management, Case study of Toyota's supply chain management in 2011

Unit-II

Insurance:

Meaning and Nature of Insurance, Role of Insurance in Risk Management, Essentials of Insurable risks, Functions of Insurance, Scope and Limitations of Insurance, Insurance and Economic Development. Fundamental Concepts -Double Insurance, Reinsurance, Over Insurance, Under Insurance, Co insurance.

Unit-III

Insurance Contract:

Essential Elements of an Insurance Contract, Fundamental Principles of Insurance contract.
Case Study : Application of Insurance Principles

Unit-IV

Indian Insurance Industry:

Historical Background of Indian Insurance Industry, Current Structure of Insurance Industry in India, Public Sector in Insurance Industry, Private Sector in Insurance Industry. Reforms and their impact on the Indian Insurance Sector.

Suggested Readings:

1. S.C.Sahoo and S.C.Dass: Insurance Management Text & Cases, Himalaya Publishing House, New Delhi.
2. George E.Rejda, Principles of Risk Management and Insurance, Tenth Ed. (2011), Pearson Education.
3. Gupta P.K., Insurance and Risk Management , Himalaya Publishing House; (2004)
4. K.C. Mishra & G.E. Thomas, General Insurance- Principles and Practices, (2009), Cengage Learning India Pvt.
5. M.N. Mishra and S.B.Mishra: Insurance: Principles and Practices, S. Chand and Sons, New Delhi.
6. Monika Mittal & Reena Matta, Insurance Management, Kalyani Publishers.
7. Insurance Institute of India, Mumbai, IC-01, Principles of Insurance.

SEMESTER-II

INSDSC2:Life Insurance

Duration: 3 Hours **Max. Marks: 90**
Internal Assessment:10

Objective: By studying this paper, the students will have an in-depth understanding of the life insurance products, policies, principles & practices followed in life insurance business. The paper will acquaint the students with the different services and products offered by life insurance business.

Unit-I

Introduction:
Overview and Meaning of Life Insurance, Importance of Life insurance in Financial Planning, Procedure of effecting a Life Insurance Policy, Procedure For Settlement of Claims, Types of Life Insurance Products and their applicability to different situations

Unit-II

Life Insurance Contract
Essential Features of a Life Insurance Contract, Fundamental Principles of Life Insurance Contract, Life Insurance Annuity Contract, Policy Provisions:Grace Period, Lapse revival, Paid up policies, Nomination, Assignment, Surrender value, Suicide clause.

Unit-III

Underwriting and Risk Classification:
Meaning and objective of Underwriting, Underwriting Process, Underwriting in Life Insurance, Factors affecting life insurance risk, Classification of Life Insurance Risk, Treatment of Sub Standard of Risk

Unit -IV

Life Insurance Business
Life Insurance Players in India, Important Life Insurance Policies issued by LIC of India, Progress of Life Insurance Business in India.

Suggested Books:
1. Huebner S.S. and Kenneth Black Jr. : Life Insurance- Prentice Hall Inc. EngleWood Clifts, New Jersy.
2. Mehar Robert L. : Life Insurance : Theory and Practice- Business Publication, Texas.
3. Meclean: Life Insurance
4.Gupta, O.S.: Life Insurance – (Frank Brothers, New Delhi).
5.Mishra, M.N. : Insurance Principles and Practice Delhi, Vikas Publishing House.

Semester-I

1	**Course Title**	RETAIL MARKETING
2	**Course Code**	-
3	**Typology of Course**	-
4	**Course Duration/ LTP** (Lectures, Tutorials, Practical)	45 hours of lectures for Theory + 15 hours of tutorials (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 credit = 15 hours of tutorials based on 1 hr./week) (1 Semester = 15 weeks)
5	**Credits**	4 Credits Theory = 4 credits
6	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) 4X5 + 4 X 15 = 80 Marks a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Small Case lets could be added.
7	**Course Objectives**	This course intends to provide the students with an overview of the retail industry, concepts & processes and an opportunity to understand the areas of accountability for a retail manager. The students will also be able to determine a level of interest in pursuing a career in retail industry.
8	**Course Outcomes**	The students who successfully complete this course will be able to: CO1: describe the theoretical and applied aspects of retail marketing. CO2: define the various methods of classifying retail formats. CO3: compare and contrast single retail channel with multi-channel retailing. CO4: recognize the challenges faced by multichannel retailers. CO5: summarize the data that could be collected to support the multichannel shopping experience of the future. CO6: understand the various retail marketing environmental issues
9	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Tutorials and Group Discussions, Case Studies etc.

10	**Instructions for paper setters and candidates**	**Note:** The question paper of each subject covering the entire course shall be divided into three sections. **Section A (20 marks):**This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 5 marks; the total weightage being 20 marks. **Section B (30 marks):**This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 15 marks; the total weightage being 30 marks. **Section C (30 marks):** This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 15 marks; the total weightage being 30 marks.
11	**Course Content**	**Unit-I** **Introduction to Retailing**- Meaning, Nature of Retailing, Types of Retailers, Theories of Retail Institutional Change. **Retailing Channels-** Multichannel Retailing- Meaning and its Types,Issues in Multichannel Retailing, Shopping Experience **Unit-II** **Introduction to Retail Marketing-** Concept of Retail Marketing; Concept of Marketing; Difference between Marketing and Selling; Marketing Management tasks; Marketing Orientation **Retail Environment-** Forces in retailers' micro environment- Suppliers, Intermediaries, Customers, Competitors; Forces in retailers macro environment- Demographic, Political/Legal, Social/Cultural, Economic, Technological; Trends in Indian Retail Industry
12	**Suggested Readings**	• J.N. Jain & P.P. Singh, Modern Retail Management, Regal Publications, New Delhi • Suja Nair, Retail Management, Himalaya Publishing house • Chetan Bajaj, Tuli & Srivastava, Retail Management, Oxford University Press, New Delhi.2010 • Fernie, Principles of Retailing, Elsevier Publishing, 2010 • Swapna Pradhan, Retail Management, Text & Cases, Tata McGraw-Hill PublishingCo, New Delhi, 2008

Semester-II

1	**Course Title**	RETAIL MARKETING
2	**Course Code**	-
3	**Typology of Course**	-
4	**Course Duration/ LTP** (Lectures, Tutorials, Practical)	45 hours of lectures for Theory + 15 hours of tutorials (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) (1 credit = 15 hours of tutorials based on 1 hr./week) (1 Semester = 15 weeks)
5	**Credits**	4 Credits Theory = 4 credits
6	**Max. Marks and Time**	100 Marks (80 TH + 20 IA) 4X5 + 4 X 15 = 80 Marks a. The exam of this course will be of 3 hours duration. b. There will be no objective type questions. c. Small Case lets could be added.
7	**Course Objectives**	This course intends to provide the students with an overview of the retail industry, concepts & processes and an opportunity to understand the areas of accountability for a retail manager. The students will also be able to determine a level of interest in pursuing a career in retail industry.
8	**Course Outcomes**	The students who successfully complete this course will be able to: CO1: identify the core concepts of marketing and understand the needs of the customer. CO2: apply the company orientation towards the market and also evaluate the market. CO3: analyze and categorize segmentation, targeting and positioning of the product and the consumer. CO4: understand what marketing means to business executives and academics. CO5: analyze the consumer behavior and can recognize the different steps in buying decision process. CO6: apply the essentials for successful implementation of service management.
9	**Pedagogy of the Course Work**	Class Room Lecture, Interactive sessions, PPT, Tutorials and Group Discussions, Case Studies etc.

10	**Instructions for paper setters and candidates**	**Note:** The question paper of each subject covering the entire course shall be divided into three sections. **Section A (20 marks):**This section will have 6 short answer questions from the entire syllabus. Students are required to attempt 4 questions from this section. Each question will carry 5 marks; the total weightage being 20 marks. **Section B (30 marks):**This section will consist of essay type/numerical questions from Unit I of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 15 marks; the total weightage being 30 marks. **Section C (30 marks):** This section will consist of essay type/numerical questions from Unit II of the syllabus. The candidate will be required to attempt two questions out of four questions. Each question will carry 15 marks; the total weightage being 30 marks.
11	**Course Content**	**Unit-I** **Consumer Behaviour and Retail Operations-** Consumer Buying Behaviour in retail; Main theories of Consumer Behaviour; Consumer Buying Decision Process for retail management; Social factors influencing Buying Process **Retail Marketing Segmentation-** Meaning of market segmentation; Importance of Market Segmentation in retail; Criteria for effective Segmentation; Basis for market segmentation **Unit-II** **Service Management in Retailing-** Meaning of service management; Characteristics of services; Classification of services; Essentials for successful implementation of service management **Retail Marketing Mix and Retail Product-** Meaning of Marketing mix; Elements of marketing mix; Breakdown of Retailing as - Retail Product, Service, Quality, Merchandise, Brand Name, Features and Benefits and Atmospherics.
12	**Suggested Readings**	• Leon G Schiffman, Joseph Wisenblit and S. Ramesh Kumar, Consumer Behaviour, Pearson, USA • Rodger D Blackwell, Paul W Miniard, James F Engel and Zillur Rahman, Consumer Behaviour, Cengage Learning India Pvt Ltd, Delhi • S.M. Jha, Servicemarketing, Himalaya Publishing House, Delhi • Steve Baron and Kim Harris, Service marketing: Text & Cases, Macmillian Business, London • Swapna Pradhan, Retail Management, Text & Cases, Tata McGraw-Hill PublishingCo, New Delhi, 2008 • Russell Winer, Marketing Management, Pearson Education, Delhi

STATISTICS

Semester-I
Paper: (STADSC-1)

Credit: 4L-3, P-1

Notes:

1. A candidate shall offer this subject in B.Sc. (Hons) only if he/she studied Mathematics as a subject at the 10+2 level.
2. Students are to be engaged for 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).
3. Practical examination will be held in the respective semester.

Paper Title with code	Lecture (L) / Credit	Practical (P) / Credit	Marks Distribution	
B.Sc. STAT-MIN-101: DESCRIPTIVE STATISTICS - I	Lecture:3 Credit:3	Practical Lecture:2 Credit: 1	Theory	Practical
			Max .Marks :75 Theory: 60 Internal Assessment:15 Time:3Hours	Max .Marks : 25 Viva voce:5 Record of the practical: 5 Written practical Exam: 15 Time: 3Hours

DESCRIPTIVE STATISTICS-I

Objective:
The objective of the course is to gain proficiency in presenting data through frequency distribution, diagrams, and graphical representations. To apply descriptive statistical techniques/methods to real-world datasets and draw meaningful insights. The focus will be both on theoretical as well as practical approach based on descriptive statistics.

Learning Outcomes: On completion of the course, the student will be able to:

1. Understand the origin, development, scope, and importance of statistics.
2. Understand the nature of statistical data.
3. Identify the need for Classification and Tabulation.
4. Understand moments and their relationships, including their effects on the change of origin and scale.
5. Construct frequency tables, interpret the data, and identify the importance of diagrammatic and graphical presentation of data.
6. Explain, evaluate, and interpret various measures of central tendency, dispersion, skewness, and kurtosis.

Note: There will be nine questions in all. Question No. 1 will be compulsory covering the whole of the syllabus and comprising 5 to 7 short answer-type questions. The rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all, selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Introduction of Statistics: Origin, development, definition, scope, uses, and limitations.

Types of Data: Qualitative and quantitative data, nominal and ordinal data, time series data, discrete and continuous data, frequency and non-frequency data.

Collection and Scrutiny of Data: Collection of primary and secondary data - its major sources including some government publications, scrutiny of data for internal consistency and detection of errors of recording, classification, and tabulation of data.

UNIT-II

Presentation of Data: Frequency distribution and cumulative frequency distribution, diagrammatic and graphical presentation of data, construction of bar, pie diagrams, histograms, frequency polygon, frequency curve, ogives, stem-and-leaf chart, Box and Whisker plot.

Measures of Central Tendency or Averages: Concepts of central tendency, characteristics for an ideal measure of central tendency, arithmetic mean, median, mode, geometric mean, harmonic mean; partition values - quartiles, deciles, percentiles, and their graphical location along with their properties, applications, merits, and demerits.

UNIT-III

Measures of Dispersion: Concept of dispersion, characteristics for an ideal measure of dispersion. Absolute and relative measures based on: range, interquartile range, quartile deviation, coefficient of quartile deviation, mean deviation, coefficient of mean deviation, variance, standard deviation (σ), coefficient of variation and properties of these measures, root mean square deviation and their relationship, variance of the combined series.

Moments: Moments about mean and about any point and derivation of their relationships, effect of change of origin and scale on moments, Sheppard's correction for moments (without derivation), Charlier's checks; Pearson's β and γ coefficients.

UNIT-IV

Skewness and Kurtosis: Concept of skewness, its types, Absolute and relative (coefficient) measures of skewness with their interpretations, and concept of Kurtosis, its types, and coefficients of kurtosis with their interpretations.

Theory of Attributes: Symbolic notations, dichotomy of data, class frequencies, order of class frequencies, consistency of data, independence and association of attributes, Yule's coefficient of association and coefficient of colligation and their relationship.

Books Recommended:

S.No.	Title of Book	Name of Author	Publisher
1	Fundamentals of Statistics Vol. I	Goon A.M., Gupta M.K., Dasgupta B.	World Press, Calcutta
2	Statistics	Johnson R.	Wiley Publishers
3	Basic Statistics	Aggarwal B.L.	New Age International
4	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons

STADSC-1: DESCRIPTIVE STATISTICS-I (PRACTICAL)

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper B.Sc. STAT-MIN-101 will be based on the following topics:

1. To construct frequency distributions using exclusive and inclusive methods.
2. Representation of data using Bar and pie diagrams.
3. Representation of data using Histogram, Frequency Polygon, Frequency Curve, and Ogives.
4. Graphs: stem-and-leaf display and Box and Whisker Plot.
5. To compute various measures of central tendency.
6. To compute various measures of dispersion.
7. To obtain first four moments for the given grouped frequency distribution.
8. To obtain various coefficients of skewness and kurtosis.
9. To discuss the association of attributes for a 2x2 contingency table using Yule's coefficient of association and colligation.

Semester-II

Paper: (STADSC-2)

Credit: 4
L-3, P-1

Notes:

1. Students are to be engaged for 5 hours of teaching per week (3 lectures for theory and 2 lectures for practical).
2. Practical examination will be held in the respective semester.

Paper Title with code	Lecture (L) / Credit	Practical (P) / Credit	Marks Distribution	
B.Sc. STAT-MIN-102 : PROBABILITY THEORY	Lecture:3 Credit:3	Practical Lecture:2 Credit:1	Theory	Practical
			Max. Marks: 75 Theory: 60 Internal Assessment:15 Time:3Hours	Max. Marks: 25 Vivavoce:5 Record of the Practical:5 Written practical Exam: 15 Time:3Hours

Probability Theory

Objective:

The main objective of the course is to learn different approaches to defining probability methods. Apply probability concepts and techniques to analyze and solve problems in various fields. Also, the aim is to familiarize with random variables, probability functions, and their distributions, including discrete and continuous cases. Gain proficiency in generating functions and understand their properties. The focus will be on a theoretical as well as practical approach.

Learning Outcomes: On completion of this course, students will be able to:

- Understand the concepts of probability, random variables, and independence of random variables.
- Explain and apply various definitions and approaches to probability including classical, relative frequency, statistical, and axiomatic approaches.
- Understand and apply moments generating function, cumulant generating function, and probability generating function.
- Utilize generating functions to compute moments and cumulants.
- Analyze functions of random variables and compute joint, marginal, and conditional probability distribution functions.
- Apply different types of probability approaches to solve problems.

Note: There will be nine questions in all. Question No. 1 will be compulsory covering the whole of the syllabus and comprising 5 to 7 short answer-type questions. The rest of the eight questions will be set from the four units uniformly i.e. two from each unit. The candidate will be required to attempt five questions in all, selecting one question from each unit and the compulsory one. All the questions will carry equal marks.

UNIT-I

Concepts in Probability: Random experiment, trial, sample point, sample space, operation of events, exhaustive, equally likely, mutually exclusive and independent events; Definition of probability - classical, relative frequency, statistical and axiomatic approach.

UNIT-II

Conditional Probability: Addition and multiplication laws of probability and their extension to events. Boole's inequality; Bayes' theorem and its applications.

UNIT-III

Random Variable and Probability Functions: Definition of random variable, discrete and continuous random variable, probability function, probability mass function and probability density functions, distribution function and its properties, functions of random variables, joint, marginal and conditional probability distribution function.

UNIT-IV

Mathematical Expectation: Definition and its properties - moments, addition and multiplication theorem of expectation. Conditional expectation and conditional variance.

Generating Functions: Moments generating function, cumulant generating function, probability generating function along with their properties. Characteristic function.

Books Recommended:

S.No.	Title of Book	Name of Author	Publisher
1	Fundamentals of Mathematical Statistics	Gupta S.C. & Kapoor V.K.	Sultan Chand & Sons
2	Elementary Probability	David S.	Oxford Press
3	Introduction to the Theory of Statistics	Mood A.M., Graybill F.A., Boes D.C.	McGraw Hill
4	Introduction to Mathematical Statistics	Kapoor & Saxena	S. Chand

STADSC-2: PROBABILITY THEORY (PRACTICAL)

Notes:

(i) The Practical Question Paper will contain five questions. A student will be required to attempt any three questions, each of 5 marks, in three hours duration.

(ii) The practical of paper B.Sc. STAT-MIN-102 will be based on the following topics:

1. Problems based on classical, statistical, and axiomatic definition of probability.
2. Problems based on addition and multiplication law of probability.
3. Problems based on conditional and Bayes' theorem of probability.
4. Problems based on univariate random variables and distribution function.
5. Problems based on marginal and conditional probability distribution function.
6. Problems based on mathematical expectation.
7. Problem based on generating functions.

Zoology

Programme Outcomes (PO)

PO1:
Students will be able to create, apply, and disseminate knowledge of zoology in basic as well as applied domains.

PO2:
Students will have the ability to identify, formulate, analyze and solve problems related to animal science and its application at both curricular and research level through critical thinking.

PO3:
Students can apply ICT based skills and they will be literate in softwares like MS-office, various Google Apps so as to use them inacademics.

PO4:
Students will be able to appreciate diversity of nature, think of environment sustainability and will have broader view of problems of pollution, deforestation and species extinction.

PO5:
Students will be able to understand different aspects of human health and vision towards a healthy society.

PO6:
Students will have enthusiasm to pursue higher studies and find placement avenues through it.

Programme Specific Outcomes (PSO)

PSO1
Student are able to apply the knowledge of core concepts of zoology in national level competitive examinations like UPSC, Forestry, Wild life, State specific civil service examinations, B.Ed./M.Sc. entrance tests.

PSO2
Students will have skill based knowledge in the fields of apiculture, fisheries, microtomy. This strengthens the students for small scale entrepreneurship.

PSO3
Students will be capable enough to make use of MS-Word and MS-PowerPoint, Web-based academic links to broaden their ICT based skills.

PSO4
Students will be able to identify pathogenic and harmful animal species for better management of agriculture, dairy, poultry as well as human health.

Semester- I

SUBJECT A

Course Code: ZOODSC1
Course Title: Diversity of Non-Chordates

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + T=0 + P=1)**
Total Marks= 100 (Theory=75 + Practical=25)
Theory Marks= 75 {60 (Theory examination) +15 (Internal assessment)}
Practical Marks= 25 {20 (Practical examination) +5 (Internal assessment)}
Duration for final examination: 3 hours

Instructions For Paper Setter and candidate

Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 8 parts of 1.5 marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

Theory

Course Outcomes:

- *The students will be able to develop an appreciation for the biodiversity of invertebrate species.*
- *The students will acquire knowledge about co-existence of different forms of living organisms ranging from unicellular to multicellular animals.*
- *The students will be able to identify different non-chordates in their surroundings.*
- *The students will be able to apply their knowledge towards identification of pathogenic non-chordates and their management.*

UNIT- 1

Protozoa: General characteristics and classification up to classes; Eco-Morphological notes on *Euglena, Amoeba*, *Paramecium &Plasmodium.*
Locomotion and reproduction in Protozoans.
Porifera: General characteristics and classification up to classes; Eco-Morphological notes on *Sycon, Hyalonema, Euplectella & Spongilla.*
Canal system and Spicules in sponges.

UNIT- 2

Cnidaria*:* General characteristics and classification up to classes; Eco-Morphological notes on *Obelia, Hydra,Physalia, Millepora, Aurelia, Tubipora, Alcyonium, Gorgonia, Metridium & Fungia.*
Polymorphism in *Cnidaria*.
Coral reefs: Types, Formation and Economic Importance.
Platyhelminthes*:* General characteristics and classification up to classes;Eco-Morphological notes on *Fasciola hepatica & Taenia solium*
Nemathelminthes*:* General characteristics and classification up to classes; Eco-Morphological notes on*Ascaris & Ancylostoma.*
Adaptations of Helminth parasites.

UNIT- 3

Annelida: General characteristics and classification up to classes; Eco-Morphological notes on *Nereis, Chaetopterus, Pheretima & Hirudinaria.*
Metamerism, Reproduction in Oligochaeta.
Arthropoda: General characteristics and classification up to classes; Eco-Morphological notes on *Limulus, Palaemon, Cancer, Scolopendra, Julus, Bombyx, Periplaneta & Palamnaeus*
Insect mouth parts, Social organization in termites and honey bee.

UNIT- 4

Mollusca: General characteristics and classification up to classes; Eco-Morphological notes on *Chiton, Dentalium, Pila, Doris, Helix, Unio, Ostrea, Pinctada, Sepia, Octopus & Nautilus.*
Torsion and Detorsion in Gastropoda, Pearl formation in Bivalves.
Echinodermata: General characteristics and classification up to classes; Eco-Morphological notes on *Asterias, Echinus, Cucumaria & Antedon.*
Water vascular system in Asteroidea.
Larval forms in Echinoderms.

_____: Diversity of Non-Chordates
Practical

1. **Protozoa:** Study of stained slides of *Euglena, Amoeba*, *Paramecium*, *Plasmodium,* Binary fission and conjugation in *Paramecium***.**
2. **Porifera:** Study of slides of *Sycon* (T.S. and L.S.); Study of Specimens: *Sycon, Hyalonema, Euplectella* & *Spongilla.*
3. **Cnidaria:** Study of slides of *Obelia, Hydra;* Study of Specimens: *Physalia, Millepora, Aurelia, Tubipora, Alcyonium, Gorgonia, Metridium & Fungia.*
4. **Platyhelminthes:** Study of Specimens: *Fasciola hepatica & Taenia solium.*
5. **Nemathelminthes:** Study of Specimens: *Ascaris & Ancylostoma*.
6. **Annelida:**Study of Specimens: *Nereis, Chaetopterus, Pheretima & Hirudinaria.*
7. **Arthropoda:**Study of Specimens: *Limulus*, *Palaemon*, *Cancer*, *Scolopendra*, *Julus*, *Bombyx, Periplaneta & Palamnaeus.*
8. **Mollusca:** Study of Specimens: *Chiton, Dentalium, Pila, Doris, Helix, Unio, Ostrea, Pinctada, Sepia, Octopus & Nautilus.*
9. **Echinodermata:** Study of Specimens: *Asterias, Echinus, Cucumaria &Antedon.*
10. **Educational trip/ Project work**.

SUGGESTED READINGS

1. Ruppert EE and Barnes RD (2006). *Invertebrate Zoology*, VIII Edition. Holt Saunders International Edition. ISBN No.- 9780721615622.
2. Barnes RSK, Calow P, Olive PJW, Golding DW and Spicer JI (2002). *The Invertebrates: A New Synthesis*, III Edition, Blackwell Science. ISBN No.- 9781444312331.
3. Barrington EJW (1979). *Invertebrate Structure and Functions,* II Edition, Thomas Nelson & Sons Ltd. ISBN No.- 9780470265031.
4. Brusca RC, Moore W, Shuster SM (2016). *Invertebrates*, III Edition, Sinauer Associates, Inc., Publishers. ISBN No.- 9781605353753.
5. Kotpal RL (2020). *Modern Text Book of Zoology: Invertebrates,* 12th edition, Rastogi Publications. ISBN No.- 9788193887554.
6. Jordan EL and Verma PS (2022). *Invertebrate Zoology,* S. Chand and Company Ltd. ISBN No.- 9789355011954.
7. Dhami PS and Dhami, JK (2022). *A Textbook of Zoology, Vol. I Non chordates*, Pradeep Publications. ISBN No.- 9789391966003.

8. Verma PS (2010). *A Manual of Practical Zoology: Invertebrates*, S. Chand and Company Ltd.ISBN No.- 9788121908290.
9. Lal SS (2017). *Practical Zoology Invertebrate,* Rastogi Publications. ISBN No.- 9789350781616.

SEMESTER- II
SUBJECT-A

Course Code:ZOODSC2
Course Title: Diversity of Chordates

Total Hours= 75 (Theory= 45 + Practical= 30) **Credits= 4 (L=3 + T=0 + P=1)**
Total Marks= 100 (Theory= 75 + Practical=25)
Theory Marks= 75 {60 (Theory examination) +15 (Internal assessment)}
Practical Marks= 25 {20 (Practical examination) +5 (Internal assessment)}
Duration for final examination: 3 hours

Instructions For Paper Setter and candidate

Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 8 parts of 1.5 marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts. To qualify for a Course, the candidate must obtain a minimum of 40% marks.

Theory

Course Outcomes:

- *Students will be able to develop an appreciation for the biodiversity of chordates.*
- *Students will acquire knowledge about origin, taxonomic classification, structure and functions of chordates.*
- *Students will understand the basic characters, advancements and adaptations of different types of vertebrates.*
- *Students will be able to apply their knowledge towards identification of different species of chordates in their surroundings.*

UNIT- 1

Introduction to Chordates
General characteristics and outline classification
Protochordata
General characteristics of Hemichordata, Urochordata, Cephalochordata and classification upto classes; *Balanoglossus, Herdmania& Amphioxus.*
Study of the larva and retrogressive metamorphosis in *Herdmania.*

UNIT- 2

Agnatha
General characteristics and classification of cyclostomes up to classes; *Petromyzon& Myxine.*
Pisces
General characteristics of Chondrichthyes and Osteichthyes, classification up to classes; *Scoliodon, Sphyrna, Pristis, Torpedo, Chimaera, Mystus, Heteropneustes, Labeo, Exocoetus, Echeneis, Anguilla, Hippocampus, Tetraodon, Diodon, Anabas&Solea*.
Migration, Osmoregulation and Parental care in fishes, Fish scales.
Amphibia

General characteristics and classification up to classes; *Ichthyophis/Ureotyphlus, Necturus, Bufo, Hyla, Alytes& Salamandra.*
Parental care in amphibians, Neoteny and Paedogenesis.

UNIT- 3

Reptilia
General characteristics and classification up to classes; *Chelone, Trionyx, Hemidactylus, Varanus, Uromastix, Chamaeleon, Ophiosaurus, Draco, Bungarus, Vipera, Naja, Hydrophis& Crocodylus.*
Poison apparatus and biting mechanism in snakes. Identification of poisonous and non-poisonous snakes.

Aves
General characteristics and classification up to classes; *Ardea, Milvus, Pavo, Tyto, Alcedo, Eudynamis& Casuarius.*
Archaeopteryx- a connecting link; Principles and aerodynamics of flight, Flight adaptations and Migration in birds.

UNIT- 4

Mammalia
General characters and classification up to classes; *Ornithorhynchus, Echidna*, *Macropus* (Kangaroo), *Loris, Macaca, Manis* (Scaly ant eater), *Hystrix* (porcupine), *Funambulus* (Squirrel), *Canis,Herpestes* (Mongoose), *Capra, Pteropus*.
Adaptive radiation with reference to locomotory appendages, Exoskeleton derivatives and dentition.

_____: Diversity of Chordates
Practical

1. Protochordata
Study of specimens: *Balanoglossus, Herdmania& Amphioxus.* Larva of *Herdmania* (slide).

2. Agnatha
Study of specimens: *Petromyzon& Myxine.*

3. Pisces
Study of specimens: *Scoliodon, Sphyrna, Pristis, Torpedo, Chimaera, Mystus, Heteropneustes, Labeo, Exocoetus, Echeneis, Anguilla, Hippocampus, Tetraodon, Diodon, Anabas&Solea*.

4. Amphibia
Study of specimens: *Ichthyophis/Ureotyphlus, Necturus, Bufo, Hyla, Alytes& Salamandra.*

5. Reptilia
Study of specimens: *Chelone, Trionyx, Hemidactylus, Varanus, Uromastix, Chamaeleon, Ophiosaurus, Draco, Bungarus, Vipera, Naja, Hydrophis& Crocodylus.*
Study of keys for the identification of poisonous and non-poisonous snakes

6. Aves
Study of specimens: *Ardea, Milvus, Pavo, Tyto, Alcedo, Eudynamis& Casuarius.*

7. Mammalia

Study of specimens: *Ornithorhynchus, Echidna*, *Macropus* (Kangaroo), *Loris, Macaca, Manis* (Scaly ant eater), *Hystrix* (porcupine), *Funambulus* (Squirrel), *Canis,Herpestes* (Mongoose), *Capra &Pteropus*.

8. Educational trip/ Project work.

SUGGESTED READINGS

1. Young JZ (2004).*The Life of Vertebrates*. III Edition. Oxford University Press. ISBN No.- 9780198571735.
2. Pough H (2012).*Vertebrate life.* VIII Edition, Pearson International. ISBN No.- 9780321600790.
3. Kotpal RL (2020). *Modern Text Book of Zoology: Vertebrates, 5*th edition, Rastogi Publications. ISBN No.- 9789350780954.
4. Jordan EL and Verma PS (2021). *Chordate Zoology,* S. Chand and Company Ltd. ISBN No.- 9788121916394.
5. Dhami PS and Dhami, JK (2022). *A Textbook of Zoology, Vol. I Chordates & Evolution*, Pradeep Publications. ISBN No.- 9789391966324.
6. Verma PS (2010). *A Manual of Practical Zoology: Chordates*, S. Chand and Company Ltd.ISBN No.- 9788121908306.
7. Lal SS (2015). *Practical Zoology Vertebrate,* Rastogi Publications. ISBN No.- 9789350780169.

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